# 1 Theoretical terms: Craig, Ramsey and the theoretician's dilemma

Theories introduce vocabulary that their data do not contain: field, entropy, gene. Hempel's theoretician's dilemma (Hempel, 1958) runs: if theoretical terms serve to systematise the observable facts, they are dispensable, by Craig's theorem (Craig, 1953) or by Ramsey's method (Ramsey, 1931); if not, they are pointless. In Solomonoff axiom induction hypotheses may use vocabulary that the data lack, so one can ask in which hypothesis classes theoretical terms change the posterior, and what pays: vocabulary, relations or entities (the third step of the ladder of **??**). Theorem 1.3 answers; Appendix A has the proofs.

## 1.1 Setting and the dilemma theorem

**Definition 1.1** (Observational restriction)**.** Let $\mathcal{L} = \mathcal{L}_O \cup \mathcal{L}_T$ be a countable language with equality and computable syntax (finite where derivations are charged), with *observational* part $\mathcal{L}_O$; $B$ may contain theoretical symbols. The *observational content* of $A \subseteq \mathrm{Sent}_{\mathcal{L}}$ is $T_O(A) := \mathrm{Cn}(A) \cap \mathrm{Sent}_{\mathcal{L}_O}$, and $D$ is *observational* if $\mathrm{dom}\, D \subseteq \mathrm{Sent}_{\mathcal{L}_O}$. Let $G_U^O := \{(w,\chi) \in G_U : \chi \in \mathrm{Sent}_{\mathcal{L}_O}\}$, $M_A^{\mathcal{L}_O} := \nu_{G_U^O}$, $W_A^{\mathcal{L}_O}(D) := \sum_e \mathbf{m}(e)\mathbf{1}[B \cup (\mathsf{W}_e \cap \mathrm{Sent}_{\mathcal{L}_O}) \text{ consistent and } \vdash \psi_D]$, and let $K_A^{\mathcal{L}_O}(D)$ be the least $K(e)$ over the same $e$.

**Definition 1.2** (Literal data and charged deduction)**.** *Literal data* are those of **??**, in the fresh-atom setting, with $\mathcal{L}_O = \{\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1, R\}$. *Charged deduction* is that of **??**: a hypothesis decides an axiom set $A_p$ within time $t$, satisfies **??**, is consistent, and derives each datum in the calculus $\mathsf{K}$ with every axiom instance, logical ones included, of size at most $h$ (reading (a1)), every nonlogical instance of size at most $h$ ((a2)), material at most $g$ ((b)), or nonlogical material at most $g$ ((b2)); the mixtures are $W_A^{a1}[t,h]$, $W_A^{a2}[t,h]$, $W_A^{b}[t,g]$, $W_A^{b2}[t,g]$. At *polynomial budgets* the degrees of membership time and budget are charged in the prior, as in Definition E.18 of our earlier report (Hänni and Claude (Anthropic), 2026); then $X$ is in the fit class iff one hypothesis is compatible with every $D_n^X$ (Lemma E.11 there).

A theory in a language containing $\mathcal{L}_O$ is *about the named objects* if it has a model whose $\{\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1\}$-reduct is the named structure $\mathfrak{T}$, and *compatible with the named structure* if it is consistent with the first-order theory $\mathrm{Th}_{\mathfrak{T}}$ of $(\mathfrak{T}; \mathtt{e}, \mathtt{s}_0, \mathtt{s}_1)$ (Definition A.1); "named object" is our formal stand-in for "observable thing".

**Theorem 1.3** (The theoretician's dilemma in Solomonoff axiom induction)**.** *Data are observational sentences; in (ii)–(iv), literal data under the readings of Definition 1.2 at polynomial budgets. In the fit-class statements of (iii), and in (iv) and (v), $B = \emptyset$.*

*(i)* In principle. *With deduction free, restricting hypotheses to the observational vocabulary changes the prior weight of every Borel class of observational contents, the predictive probability of every observational sentence and the cumulative loss by at most a constant, in the one-tape, coin-free and Kolmogorov versions, and also when the time to check or to generate axioms is penalised (under* **????***). Tape by tape, the observational projection of a random theory has exactly the same law on observational data.*

*(ii)* Craig's construction in practice. *When derivations are charged, Craig's axiomatisation of a theory whose observational consequences have derivations of size* ds *needs derivations of*

*size at least* $2^{\mathrm{ds}}$. *Every axiomatisation by padded observational literals with polynomial-time membership fits exactly the polynomial-time decidable regularities, under every reading. Axiom sets with theoretical symbols fit* $\mathsf{NP} \cap \mathsf{coNP}$ *(total material),* $\mathsf{EXPTIME}$ *(every instance short) and every decidable regularity (only the theory's own axioms short); so Craig's construction loses unconditionally under the last two readings and, under the first, iff* $\mathsf{NP} \cap \mathsf{coNP} \neq \mathsf{P}$.

*(iii)* Eliminating the vocabulary in practice. *Theoretical vocabulary can be eliminated at polynomial cost: per hypothesis, with* $O(1)$ *bits and a polynomial change of budget, under total material, by storing proofs instead of run times; and at the level of fit classes under (a1), (a2), (b2), by storing the eliminated relations in objects that no observational term names. Observational hypotheses, and already finite observational theories, fit what all hypotheses fit.*

*(iv)* What is indispensable. *A finite theory in the observational vocabulary with any unary theoretical predicates and constants, compatible with the first-order theory of the named objects, fits exactly the regular regularities, under every reading and also with free deduction. One binary theoretical relation over the named objects restores every decidable regularity ((a2), (b2)),* $\mathsf{NP} \cap \mathsf{coNP}$ *((b)) and* $\mathsf{EXPTIME}$ *((a1)). So does quantifying over unnamed objects; but a finite theory that does so without theoretical vocabulary, and fits a non-regular regularity, must prove some* $\{\mathsf{e}, \mathtt{s}_0, \mathtt{s}_1\}$*-sentence false in the named structure. For the theory built in Theorem A.14 that sentence is* $\exists z\, \mathtt{s}_1\mathtt{s}_1 z = z$*, true of no named object.*

*(v)* Free deduction, finite theories. *With deduction free, finite theories gain without bound from a theoretical function (infinity), from theoretical sorts (the theorems of* $\mathsf{PA}$*) and from a binary relation over the named objects; and r.e. axiom induction is finite-theory induction with theoretical sorts up to* $O(1)$*, coin-free.*

*Proof.* (i) is Theorem A.3. (ii) is Lemma A.5, Theorem A.6, and Corollary A.8, given **??**, the time hierarchy theorem and $\mathsf{APSPACE} = \mathsf{EXPTIME}$ (Chandra et al., 1981). (iii) is Propositions A.9 and A.11 and Theorem A.14, under **??**, given the one-tape simulation and $\mathsf{APSPACE} = \mathsf{EXPTIME}$. (iv) is Theorems A.14, A.16 and A.17 and Proposition A.20, given the Ehrenfeucht–Fraïssé theorem, the one-tape simulation and, for (a1), $\mathsf{APSPACE} = \mathsf{EXPTIME}$. (v) is Theorems A.28, A.31 and A.33 and Corollary A.19; the clause on $\mathsf{PA}$ is given Ryll-Nardzewski's theorem and Con($\mathsf{PA}$), the clause on sorts the $\Sigma_1$-completeness of $\mathsf{Q}$. □

## 1.2 In principle

Part (i) is Theorem A.3, which states it for each version named in (i) and each Borel class of observational contents, and proves the identity of laws tape by tape.

*Remark* 1.4 (Inductive systematisation in the mixture). By Theorem A.3, the posterior predictive probability of every observational sentence, given observational data, is the same up to a factor $2^{\pm 2c}$, $c$ the constant of that theorem, whether or not hypotheses may use theoretical vocabulary. So in the universal mixture, with deduction free, theoretical terms are not needed for inductive inference among observational sentences. This is a statement about the mixture, not about a single theory (Section 1.6).

### 1.3 In practice: Craig's construction and better eliminations

Craig's construction pads each consequence with the time at which it was found: the literal $\varphi$ found after $k$ steps becomes the axiom $\varphi^{\wedge(k+1)}$, a conjunction of $k+1$ copies. When the time is that of canonical proof search, on literal data every derivation from the Craig set has a line of size at least $2^{\mathrm{ds}}|\varphi^b|$, ds the size of a shortest derivation of $\varphi^b$ from the original theory (Lemma A.5). For a seven-sentence Horn theory, in a small proof system for ground literals and at a word of length 24, a computation from closed forms gives a Craig axiom of more than $10^{204.6}$ symbols, against 795 for an axiom that carries a certificate (Example A.7). At polynomial budgets, every *run-padded* axiomatisation (each member a power $\psi^{\wedge m}$ of a literal $\psi \in \{\varphi_w, \neg\varphi_w\}$) with polynomial-time membership fits exactly the polynomial-time regularities, under every reading (Theorem A.6); so the classical reply holds of Craig's construction, and we did not find this complexity-theoretic form of it in the literature.

Vocabulary itself can be eliminated at polynomial cost. Under reading (b), the certificate axioms $\theta_\gamma \to \varphi^b$ of **??**, observational because $\theta_\gamma$ uses only logical symbols, replace derivations hypothesis by hypothesis at $O(1)$ bits and a polynomial change of budget (Proposition A.9). Under the other readings the content goes into objects that no observational term names: in a structure of words and formal pairs, the pairs being the elements $z$ with $\mathtt{s}_1\mathtt{s}_1 z = z$, an observational formula $\theta_2(a, b)$, with a fresh bound variable $y_\theta$, says that $R$ holds of the pair of $a$ and $b$, and it can replace the atoms of a binary theoretical relation (Lemma A.10 and Proposition A.11). With empty background, for every *verifier* $\mathcal{V}$, a halting machine that accepts or rejects pairs (word, certificate) consistently, there are a finite theory in $\mathcal{L}_O \cup \{\mathsf{C}\}$, $\mathsf{C}$ binary, about the named objects and a finite set of $\mathcal{L}_O$-sentences that prove exactly the literals that $\mathcal{V}$ certifies, by derivations whose length and line size are linear in the time and space of a one-tape simulation of $\mathcal{V}$ plus the lengths of the input and the certificate; with a variant for alternating machines, finite theories of either kind fit what all hypotheses fit, under every reading (Theorem A.14). With deduction free the existence part is classical: as stated in Pakhomov and Visser (2019), every r.e. theory without finite models has a finitely axiomatised conservative extension with one more binary relation symbol (Kleene, 1952), even one into which every model expands (Craig and Vaught, 1958). To our knowledge the derivation bounds, hence the fit classes under charged deduction, and the observational translation are new.

### 1.4 Vocabulary or ontology

Without objects outside the named structure unary theoretical vocabulary does not help: with empty background, a finite theory in $\mathcal{L}_O$ plus unary predicates and constants, compatible with the named structure, that proves the literals of $X$ forces $X$ to be regular (Theorem A.16, given the Ehrenfeucht–Fraïssé theorem), and every regular $X$ has such a theory (Theorem A.17); one binary relation over the named objects, or quantification over unnamed objects in $\mathcal{L}_O$ alone, lets finite theories fit what all finite theories fit, and with deduction free the contrast holds for complexities too (Corollaries A.18 and A.19). The second way has a price: a finite theory without theoretical vocabulary that fits a non-regular regularity proves some $\{\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1\}$-sentence false in the named structure (for the observational theory built above, $\exists z\, \mathtt{s}_1\mathtt{s}_1 z = z$; Proposition A.20), while the theory with the binary relation is conservative over the theory of the named objects plus the data (Proposition A.21). One binary observable relation breaks Theorem A.16 (Proposition A.22).

### 1.5 Finite theories with free deduction

With deduction free and $B = \emptyset$, finite theories, the single sentences of $K_{\text{fin}}$ and $W_{\text{fin}}$ (**??**), can gain from theoretical vocabulary without bound. On the data "there are at least $n$ things", the least complexity of a consistent sentence in pure equality plus unary predicates that proves the data is $K^{\text{pad}}(n) \pm c$, which tends to infinity, though more slowly than every computable unbounded nondecreasing function, while with a unary function symbol $f$ the sentence saying that $f$ is injective and not surjective gives $O(1)$ (Theorem A.28). Its Ramsey sentence is finite but second-order, the observational axiomatisation of its $\{=\}$-consequences is first-order but infinite, and no finite first-order theory in pure equality, or in pure equality with unary predicates added, has the same consequences (Proposition A.35). On the theorems of PA as data, finite $\mathcal{L}_{\text{ar}}$-theories diverge, given the theorem of Ryll-Nardzewski (1952), while theoretical sorts give bounded complexity (Theorem A.31); and for finite $\mathcal{L}$, r.e. axiom induction is, up to a constant and coin-free, finite-theory induction in a three-sorted extension with numbers, assignments and satisfaction (Theorem A.33), an effective, many-sorted form of the theorem of Kleene (1952) and Craig and Vaught (1958) of which only the Bayesian consequence is new. The equivalence is not one-tape: on the lottery stream of **??**, random r.e. theories have bounded loss and random finite theories with any theoretical apparatus do not (Remark A.34).

A concept can also pay by factoring inferences.[1] If each of $n_A$ unary predicates implies each of $n_B$ others, a theory of atom-to-atom implications needs $n_A n_B$ members, with a new intermediate predicate exactly $\min(n_A n_B, n_A + n_B)$; so the predicate pays in this format iff $(n_A - 1)(n_B - 1) > 1$, and in full first-order logic only by a constant factor (Proposition A.36). This is the elementary instance, with the exact gain, of the compression argument for predicate invention (cf. Muggleton and Buntine, 1988).

### 1.6 What this does and does not show

What bounded deduction and finiteness make indispensable is theoretical *relations* or theoretical *entities* (for the theory we build, objects outside the named structure), not vocabulary ((iii), (iv)); and in the idealised setting nothing theoretical is indispensable, because the r.e. prior already is a prior over finite theories with theoretical vocabulary and entities ((i), (v)). The results concern one poor observational language and literal data.

**Hempel's replies.** *Impracticality*: Craig's replacement axiomatisation is unmanageable. This is a theorem, but only about Craig's construction and run-padded axiomatisations under charged derivations; vocabulary as such is eliminable at polynomial cost (Theorem 1.3(ii), (iii)). *Inductive systematisation*: Hempel argued that theoretical terms may be needed for inductive, not deductive, systematisation (Hempel, 1958); the debate is surveyed by Niiniluoto (1972). Solomonoff axiom induction is an inductive systematisation, and in its universal mixture with free deduction the reply fails: theoretical terms add no inductive power among observational sentences beyond a constant factor (Remark 1.4). This is a statement about the mixture; Hempel's own notion is relative to a single theory, and our statement is not a reconstruction of it. The reply survives, in corrected form, where something theoretical pays: for finite theories and under charged derivations, and there by relations or entities, not by vocabulary.

[1] The idea of introducing a concept that factors a complete bipartite pattern of inferences first appeared in Hänni's working notes (Hänni, 2026).

# A Proofs for theoretical terms

Throughout, $B$ is decidable and consistent, "consistent" means consistent with $B$, and Cn has $B$ built in (**??**). For a sentence $\psi$, $\psi^{\wedge 1} := \psi$ and $\psi^{\wedge(m+1)} := \psi^{\wedge m} \wedge \psi$, where $\alpha \wedge \beta$ abbreviates $\neg(\alpha \to \neg\beta)$ in the syntax of $\mathsf{K}$; so $|\psi^{\wedge m}| \geq m|\psi|$. On literal data (Definition 1.2) the literal setting of **??** is assumed: $R$ does not occur in $B$, and some model of $B$ interprets the terms $t_w$ as pairwise distinct elements. Sentence length is the symbol count. The computations quoted below check constructions and counts on finite fragments; the theorems are proved in the text, and the code is in the accompanying repository.

**Definition A.1** (The named objects)**.** $\mathrm{Th}_{\mathfrak{T}}$ is the first-order theory of the named structure $(\mathfrak{T}; \mathtt{e}, \mathtt{s}_0, \mathtt{s}_1)$. A theory in a language containing $\mathcal{L}_O$ is *about the named objects* if it has a model whose $\{\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1\}$-reduct is $\mathfrak{T}$, and *compatible with the named structure* if it is consistent with $\mathrm{Th}_{\mathfrak{T}}$.

The second notion allows objects other than the named ones, but they obey every first-order law of the named objects (for instance, $\mathtt{s}_0, \mathtt{s}_1$ are injective with disjoint ranges that miss $\mathtt{e}$, and $\mathtt{s}_u x \neq x$ for every nonempty word $u$). "Named object" is our formal stand-in for "observable thing".

## A.1 In principle

**??** showed that new vocabulary changes old-vocabulary weights by a constant factor. Part (i) of Theorem 1.3 needs more: classes of contents, a theoretical background, the penalised versions, and the identity tape by tape.

**Definition A.2** (Class-restricted observational laws)**.** For a Borel set $\Theta$ of observational contents (subsets of $\mathrm{Sent}_{\mathcal{L}_O}$, product topology), $\mathcal{C}_\Theta := \{T : T \cap \mathrm{Sent}_{\mathcal{L}_O} \in \Theta\}$, and $\mathbb{U}_A(\mathcal{C}_\Theta; D)$, $M_A^{\mathcal{L}_O}(\mathcal{C}_\Theta; D)$, $W_A(\mathcal{C}_\Theta; D)$, $W_A^{\mathcal{L}_O}(\mathcal{C}_\Theta; D)$ are the class-restricted laws and weights (**??**), with $M_A^{\mathcal{L}_O}$ and $W_A^{\mathcal{L}_O}$ as in Definition 1.1. Observational restrictions of the penalised versions of **??** are their sub-sums over hypotheses whose axioms are $\mathcal{L}_O$-sentences on every tape.

**Theorem A.3** (Theoretical vocabulary costs $O(1)$ bits, class by class)**.** *There is $c$, depending on $V$, the coding, $B$ and $\mathcal{L}_O \subseteq \mathcal{L}$, such that for every finite observational $D$ and every Borel set $\Theta$ of observational contents: (a) $2^{-c}\mathbb{U}_A(\mathcal{C}_\Theta; D) \leq M_A^{\mathcal{L}_O}(\mathcal{C}_\Theta; D) \leq 2^c \mathbb{U}_A(\mathcal{C}_\Theta; D)$; (b) the same holds for $W_A(\mathcal{C}_\Theta; D)$ and $W_A^{\mathcal{L}_O}(\mathcal{C}_\Theta; D)$, and $|K_A^{\mathcal{L}_O}(D) - K_A(D)| \leq c$; (c) for each time-constructible $t(m) \geq m^{a^*}$ there is $c_t$ such that, class by class, the observational restrictions of $W_A^{\mathrm{chk},t}$ and $W_A^{\mathrm{gen},t}$ (all forms) agree with $W_A$ within $2^{\pm c_t}$ under **??**, and those of $M_A^{\mathrm{chk},t}$ and $M_A^{\mathrm{gen},t}$ agree with $M_A$ within $2^{\pm c_t}$ under **??**; (d) for every theory machine $G$ in $\mathcal{L}$, $G^O := \{(w, \varphi) : \varphi \in \mathrm{Sent}_{\mathcal{L}_O},\ B \cup A_w^G \vdash \varphi\}$ is a theory machine emitting only $\mathcal{L}_O$-sentences, and on every tape $\omega$: $A_\omega^{G^O} = T_O(A_\omega^G)$; $B \cup A_\omega^{G^O}$ is consistent iff $B \cup A_\omega^G$ is; and $B \cup A_\omega^{G^O} \vdash \varphi^b$ iff $B \cup A_\omega^G \vdash \varphi^b$, for all observational $\varphi$ and $b \in \{0, 1\}$. Hence, for every set $\Theta$, the events defining $\nu_{G^O}(\mathcal{C}_\Theta; D)$ and $\nu_G(\mathcal{C}_\Theta; D)$ coincide tape by tape, and the two laws are equal.*
*Consequently, in each version, posterior odds between two classes of observational contents and predictive probabilities of observational sentences agree within $2^{\pm 2c}$, and cumulative semimeasure losses on observational reveal sequences within $2c$.*

In outline: (d) is compactness; (a) applies (d) to $G_U$ and the mixture identity of **??** both ways; (b) maps $e$ to indices of $T_O(\mathsf{W}_e)$ and of $\mathsf{W}_e \cap \mathrm{Sent}_{\mathcal{L}_O}$; (c) pads the observational content as in **??**, the only place where Craig's padding enters. $G^O$ is the pull-back machine of **??** for the inclusion $\mathrm{Sent}_{\mathcal{L}_O} \subseteq \mathrm{Sent}_{\mathcal{L}}$.

*Proof.* (d) $G^O$ is c.e. because $G$ is and derivability from the decidable $B$ is c.e. By compactness, and because $A^G_w$ grows with $w$ along the prefixes of $\omega$ (**??**(a)), $A^{G^O}_\omega = \bigcup_{w\sqsubset\omega}\{\varphi \in \mathrm{Sent}_{\mathcal{L}_O} : B\cup A^G_w \vdash \varphi\} = T_O(A^G_\omega)$. Since $T_O(A) \subseteq \mathrm{Cn}(A)$, consistency of $B \cup A$ passes to $B \cup T_O(A)$; if $B \cup A$ is inconsistent, $T_O(A)$ contains the observational sentence $\neg\forall x\, x = x$. Again from $T_O(A) \subseteq \mathrm{Cn}(A)$, $\mathrm{Cn}(T_O(A)) \cap \mathrm{Sent}_{\mathcal{L}_O} = T_O(A)$, so an observational $\varphi^b$ follows from $B \cup T_O(A)$ iff it lies in $T_O(A)$ iff it follows from $B \cup A$. The observational content of $G^O$ on $\omega$ is $T_O(A^G_\omega)$ itself, so membership in $\mathcal{C}_\Theta$ is preserved, and the two events coincide tape by tape.

(a) By the mixture identity of **??**, on the cylinder of a program $p$ with $V(p)$ an index of a machine $G$, the universal machine emits $A^G_{\omega'}$ on the rest $\omega'$ of the tape, hence $G^O_U$ emits $A^G_{\omega'} \cap \mathrm{Sent}_{\mathcal{L}_O}$, and $\mathbb{U}(\,\cdot \cap [p]) = 2^{-|p|}\mathbb{U}(\cdot)$ on the shifted event; here the measurability of $\mathcal{C}_\Theta$ is used (the map $\omega \mapsto T_O(A_\omega)$ is Borel, since each coordinate "$\varphi \in T_O(A_\omega)$" is the effectively open event $O_\varphi$). *Lower bound.* Apply (d) to $G := G_U$. $(G_U)^O$ is a fixed machine emitting only $\mathcal{L}_O$-sentences, so on the tapes $p\omega'$ with $V(p)$ an index of $(G_U)^O$ the filtered machine $G^O_U$ emits exactly $A^{(G_U)^O}_{\omega'}$; hence $M^{\mathcal{L}_O}_A(\mathcal{C}_\Theta; D) \geq 2^{-K((G_U)^O)}\nu_{(G_U)^O}(\mathcal{C}_\Theta; D) = 2^{-c}\mathbb{U}_A(\mathcal{C}_\Theta; D)$. *Upper bound.* $G^O_U$ is a theory machine of $\mathcal{L}$, so the same identity in $\mathcal{L}$ gives $\mathbb{U}_A(\mathcal{C}_\Theta; D) \geq 2^{-K(G^O_U)}\nu_{G^O_U}(\mathcal{C}_\Theta; D) = 2^{-c}M^{\mathcal{L}_O}_A(\mathcal{C}_\Theta; D)$. The program-less tapes contribute $(1-\Omega_V)\mathbf{1}[B \vdash \psi_D,\ T_O(\emptyset) \in \Theta]$ to both sides.

(b) Let $o(e)$ be an index of $\{\varphi \in \mathrm{Sent}_{\mathcal{L}_O} : B \cup \mathsf{W}_e \vdash \varphi\}$ and $f(e)$ one of $\mathsf{W}_e \cap \mathrm{Sent}_{\mathcal{L}_O}$; both maps are computable. By the argument of (d), $o(e)$, read in $W^{\mathcal{L}_O}_A$, is compatible with $D$ and has content in $\Theta$ iff $e$, read in $W_A$, is; and $f(e)$ read in $W_A$ behaves as $e$ read in $W^{\mathcal{L}_O}_A$. The pushforward of $\mathbf{m}$ under a computable map is a lower semicomputable semimeasure, hence at most $2^c\mathbf{m}$. So, with $P$ the compatibility-and-class predicate, $W_A(\mathcal{C}_\Theta; D) = \sum_e \mathbf{m}(e)\mathbf{1}[P(o(e))] \leq 2^c\sum_j \mathbf{m}(j)\mathbf{1}[P(j)] = 2^c W^{\mathcal{L}_O}_A(\mathcal{C}_\Theta; D)$, and symmetrically with $f$. Since $K(o(e)) \leq K(e) + c$ and $K(f(e)) \leq K(e) + c$, the statement on $K_A$ follows.

(c) *Coin-free.* For an index $e$, let $f^O_e$ be proof search from $B \cup \mathsf{W}_e$ restricted to observational inputs: accept $\varphi$ if $B \cup \mathsf{W}_e \vdash \varphi$, reject if $B \cup \mathsf{W}_e \vdash \neg\varphi$, abstain on non-observational inputs. Then $\Gamma_{f^O_e} = T_O(\mathsf{W}_e)$, which is consistent iff $\mathsf{W}_e$ is. Theorem 6.14 of our earlier report (Hänni and Claude (Anthropic), 2026), which is **??**(b), maps every consistent labeller $f$, injectively and at length cost $c_t$, to the decider (respectively enumerator) of the padded set of Lemma E.12 there, whose members are conjunction-powers of members of $\Gamma_f$, with the same consequences as $\Gamma_f$ and membership (respectively generation) within $t$, under **??**. For $f = f^O_e$ the padded set is observational and has the observational content of $\mathsf{W}_e$. Summing over $e$ with the pushforward bound gives the lower bound. The upper bound holds because the observational restriction is a sub-sum of $W^{\mathrm{chk},t}_A$ (respectively $W^{\mathrm{gen},t}_A$), and a penalised hypothesis is an ordinary one with the same theory (**??** for the class form). *One tape.* The padded random decider in the proof of **??**(a), applied to the random decider program that says yes on $\chi$ iff $\chi$ is observational and $B \cup A^G_\omega \vdash \chi$ (dovetailed proof search over the tape), has on every tape the padded copy of $T_O(A^G_\omega)$ as its yes-set; this set is observational, has the observational content of $A^G_\omega$, and meets the time bound uniformly in $\omega$ under **??**. The program-less tapes are matched by the decider of $\emptyset$, as in **??**. Sum as in (a). For the upper bound, a random decider or enumerator is read as a theory machine through a constant-size wrapper, and **??** bounds the penalised mixture by $M_A$ class by class.

*The consequences.* Divide the class bounds for $\Theta_1$ and $\Theta_2$, and the bounds for $D + (\varphi, b)$ and $D$; the semimeasure loss is $-\log M(D_n) + \log M(\emptyset)$ for each version $M$. □

A direct computation on a propositional analogue (three observational atoms, two theoretical atoms, the theoretical background $(t_0 \to o_0) \wedge (t_1 \to \neg o_2)$, 40 random theory machines on 16 tapes) finds the law of $G^O$ equal to that of $G$ on all 24480 pairs of machine and observational

data set and on 21830 class checks, while dropping the theoretical axioms instead of projecting changes the law of 37 of the 40 machines; an independent implementation with three backgrounds agrees.

## A.2 Craig's construction

**Lemma A.4** (Re-checking a literal)**.** *On literal data, let $A$ be a set of sentences each logically equivalent to a literal $\pm\varphi_{w'}$, with $B \cup A$ consistent. If $B \cup F \vdash \varphi_w^b$ for a finite $F \subseteq A$, then $F$ contains a member equivalent to $\varphi_w^b$.*

*Proof.* Suppose not. The members of $F$ are equivalent to literals; no two of them are complementary, since $B \cup F$ is consistent; and those about $w$, if any, are equivalent to $\varphi_w^{1-b}$. Take a model of $B$ in which the $t_{w'}$ are pairwise distinct and reinterpret $R$, which does not occur in $B$: put $t_{w'}$ into $R$ iff $R(t_{w'})$ is among the literals of $F$, and in addition put $t_w$ into $R$ iff $b = 0$. This model satisfies $B$, every member of $F$ and $\neg\varphi_w^b$. □

**Lemma A.5** (Craig's padding writes a run time in unary)**.** *Let $B \cup T$ be consistent.* Canonical proof search $f_T^{\mathrm{can}}$ *tests, on input $\varphi$, the binary strings in length-lexicographic order as codes of* $\mathsf{K}$*-derivations from $B \cup T$, in a code in which a derivation of symbol size $s$ has code length at least $s$, spending at least one step per string, and stops at the first derivation of $\varphi$ or of $\neg\varphi$. Let* $\mathrm{ds}(\varphi^b)$ *be the least symbol size of a derivation of whichever of $\varphi, \neg\varphi$ is provable from $B \cup T$. (a) The running time of $f_T^{\mathrm{can}}$ on $\varphi$ is at least $2^{\mathrm{ds}(\varphi^b)} - 1$. (b) On literal data, let $A^{\mathrm{Cr}} := \{(\varphi_w^b)^{\wedge(k+1)} :$ $f_T^{\mathrm{can}}$ decides $\varphi_w$ with $b$ in exactly $k$ steps$\}$ be Craig's set of $f_T^{\mathrm{can}}$ restricted to literal inputs, or its version padded by a work counter (Lemma E.12 of the earlier report). Every* $\mathsf{K}$*-derivation of $\varphi_w^b$ from $B \cup A^{\mathrm{Cr}}$ contains a line of size at least $2^{\mathrm{ds}(\varphi_w^b)}|\varphi_w^b|$; so its size, its material and its largest axiom instance are all at least that. (c) With Craig's original padding by the Gödel number* gn *of a derivation (the number whose binary expansion is* 1 *followed by the code), $A^{\mathrm{gn}} := \{(\varphi_w^b)^{\wedge \mathrm{gn}}\}$, the same bound holds.*

*Proof.* (a) Every derivation of $\varphi^b$ has size at least ds, hence code length at least ds, and no derivation of $\varphi^{1-b}$ exists. So all $2^0 + 2^1 + \cdots + 2^{\mathrm{ds}-1} = 2^{\mathrm{ds}} - 1$ strings of length less than ds are tested and fail. (b) By Lemma A.4 the derivation cites a member of $A^{\mathrm{Cr}}$ equivalent to $\varphi_w^b$. The only such member is $(\varphi_w^b)^{\wedge(k+1)}$, with $k \geq 2^{\mathrm{ds}} - 1$ by (a); for the padded version the counter is at least the simulated time. That member is a line of the derivation, of size at least $(k+1)|\varphi_w^b|$. (c) Every member of $A^{\mathrm{gn}}$ equivalent to $\varphi_w^b$ is a power with $\mathrm{gn} \geq 2^\ell$ factors, where $\ell \geq \mathrm{ds}$ is the code length; apply Lemma A.4. □

**Theorem A.6** (Run-padded axiomatisations fit exactly $\mathsf{P}$)**.** *On literal data, with run-padded sets as in Section 1.3: (a) if $A$ is run-padded, with membership decidable in polynomial time, and is compatible with every $D_n^X$ under reading (a1), (a2), (b) or (b2) with a polynomial budget $h$, or with derivations of size at most $h$, then $X \in \mathsf{P}$; (b) every $X \in \mathsf{P}$ is fitted under all these readings, at polynomial budgets with polynomial-time membership, by a run-padded set.*

*Proof.* (a) Under each reading, the member of $A$ that a derivation of $\varphi_w^b$ cites and that is equivalent to $\varphi_w^b$ (Lemma A.4) has size at most $h(|\varphi_w|)$: it is an axiom line, a nonlogical instance, a part of the material, and a line of a derivation of size at most $h$. Its form is $(\varphi_w^b)^{\wedge m}$ with $m \leq h(|\varphi_w|)$. Decide $w$ as follows: for $m = 1, \ldots, h(|w| + 2)$ test whether $(\varphi_w)^{\wedge m} \in A$, and accept iff one is. These are polynomially many tests of polynomial time. If a power of $\varphi_w$ is in $A$, then $B \cup A \vdash \varphi_w$, so $w \in X$ (otherwise $B \cup A$ would also prove the datum $\neg\varphi_w$ and be inconsistent). If $w \in X$, the

derivation within budget cites such a power with $m \le h$. (b) Clock a polynomial-time decider $f$ of $X$ at $\tau(m) := (m+2)^k$. Its Craig set $\{(\varphi_w^b)^{\wedge(j+1)} : f \text{ decides } \varphi_w \text{ with } b \text{ in exactly } j \text{ steps}\}$ is parameter-free, has polynomial-time membership (read the word off a candidate with $m$ factors and run $f$ for $m-1$ steps), and derives each datum by citing $(\varphi_w^b)^{\wedge(j+1)}$, one substitution instance of a fixed propositional derivation of $(\beta \wedge \gamma) \to \gamma$, and modus ponens; size, material and largest instance are $O((\tau(m)+1)(m+4))$. This is Lemma E.22 of the earlier report. □

A run-padded axiom carries $O(\log)$ bits, its length, while the axiom $\theta_\gamma \to \varphi^b$ of **??** carries the bits of a certificate $\gamma$. So the natural strengthening of Theorem A.6(a) to all sets whose members are each equivalent to a literal fails: certificate sets are such sets and fit $\mathsf{NP} \cap \mathsf{coNP}$ under (b). Theorem A.6 does not contradict the standard fact that every r.e. theory has an axiomatisation decidable in polynomial time, by a variant of Craig's construction (Krajíček, 2025): it concerns what such sets fit when derivations are charged. Against all hypotheses (**??**), run-padded sets lose unconditionally under (a1), (a2), (b2), and under (b) iff $\mathsf{NP} \cap \mathsf{coNP} \neq \mathsf{P}$ (Corollary A.8).

**Example A.7** (The size of a Craig axiom)**.** Take the seven-sentence Horn theory with theoretical predicates Ev, Od (an even, respectively odd, number of 1s): $\mathrm{Ev}(\mathtt{e})$; $\mathrm{Ev}(x) \to \mathrm{Ev}(\mathtt{s}_0 x)$; $\mathrm{Ev}(x) \to \mathrm{Od}(\mathtt{s}_1 x)$; $\mathrm{Od}(x) \to \mathrm{Od}(\mathtt{s}_0 x)$; $\mathrm{Od}(x) \to \mathrm{Ev}(\mathtt{s}_1 x)$; $\mathrm{Ev}(x) \to R(x)$; $\mathrm{Od}(x) \to \neg R(x)$. In a small proof system, a derivation is a sequence of ground literals over $\mathrm{Ev}, \mathrm{Od}, R$ with terms $t_u$, $|u| \le |w|$, each justified by a rule whose premise occurs earlier; a literal has $|u|+2$ symbols, one more if negated; canonical search tests sequences by number of lines, then lexicographically. For a word $w$ of length $n$ there are $N_{\mathrm{cand}} = 4(2^{n+1}-1)$ candidate lines; a shortest derivation has $L^* = n+2$ lines ($\mathrm{Ev}(\mathtt{e})$, the parity atoms of the $n$ nonempty suffixes of $w$, and the literal), and none is shorter, since each rule application extends the term by one letter and needs its premise earlier; its size is $\mathrm{ds}_T = \sum_{j=0}^{n}(j+2) + (n+2) + \mathbf{1}[\#_1 w \text{ odd}]$; and the number $k$ of sequences tested is at least $\sum_{L=1}^{L^*-1} N_{\mathrm{cand}}^L$, since every sequence with fewer lines is tested and fails. The Craig axiom has at least $(k+1)|\varphi^b|$ symbols; the certificate axiom $\theta_\gamma \to \varphi^b$, with $\gamma$ coding the shortest derivation in $L^*(2+\lceil \log N_{\mathrm{cand}} \rceil)$ bits, has $|\gamma|+15+|\varphi^b|$; the Craig axiom of a fast parity labeller has $(n+3)|\varphi^b|$.

At $n = 24$ these closed forms give $N_{\mathrm{cand}} = 134217724$, $L^* = 26$, $\mathrm{ds}_T = 376$, a Craig axiom of more than $10^{204.6}$ symbols, and 795 and 702 symbols for the certificate axiom and for the fast labeller's Craig axiom (a direct computation, which also reproduces, for $|w| \le 4$, an exact count by exhaustive search; code in the accompanying repository).

**Corollary A.8** (The practical horn)**.** *Under* **??***, on literal sequences, at polynomial budgets with polynomial membership and degrees charged, the fit classes are:*

| *hypotheses* | *(b)* | *(a1)* | *(a2), (b2)* |
|---|---|---|---|
| *run-padded observational sets (Craig type)* | $\mathsf{P}$ | $\mathsf{P}$ | $\mathsf{P}$ |
| *observational certificate sets* | $\supseteq \mathsf{NP} \cap \mathsf{coNP}$ | $\supseteq \mathsf{NP} \cap \mathsf{coNP}$ | $\supseteq \mathsf{NP} \cap \mathsf{coNP}$ |
| *all observational hypotheses*$^\dagger$ | $\mathsf{NP} \cap \mathsf{coNP}$ | $\mathsf{EXPTIME}$ | *all decidable* |
| *finite theories in* $\mathcal{L}_O$*, unary predicates, constants*$^{*\dagger}$ | *regular* | *regular* | *regular* |
| *all hypotheses* | $\mathsf{NP} \cap \mathsf{coNP}$ | $\mathsf{EXPTIME}$ | *all decidable* |

*(*$^*$*compatible with the named structure;* $^\dagger$*with* $B = \emptyset$*.) Hence under (a1) the run-padded sets fit strictly less than the axiom sets with theoretical symbols and, with* $B = \emptyset$*, than the observational hypotheses that are not run-padded, unconditionally; under (a2) and (b2) also unconditionally; and under (b) iff* $\mathsf{NP} \cap \mathsf{coNP} \neq \mathsf{P}$*.*

*Proof.* Row 1 is Theorem A.6. Row 2: the certificate sentence $\theta_\gamma := Z \to (C_\gamma \to Z)$ of **??**, with $Z$ a closed tautology and $C_\gamma$ a valid sentence coding $\gamma$, uses only logical symbols, so the certificate sets $\{\theta_\gamma \to \varphi_w^b\}$ are observational; they fit $\mathsf{NP} \cap \mathsf{coNP}$ under every reading by the third inequality of **??**(b) (material $2|\gamma| + 29 + |\varphi^b|$, largest instance $|\gamma| + 15 + |\varphi^b|$, every nonlogical instance a member). Row 3 is Proposition A.9 for (b), Proposition A.11 for (a1) and Theorem A.14 for (a2), (b2), each bounded above by row 5. Row 4 is Theorems A.16 and A.17. Row 5 is **??**. The separations: $\mathsf{P} \subsetneq \mathsf{EXPTIME}$ by the time hierarchy theorem; some decidable set is not in $\mathsf{P}$; and under (b) the classes are $\mathsf{P}$ and $\mathsf{NP} \cap \mathsf{coNP}$. □

## A.3 Eliminating vocabulary

**Proposition A.9** (Elimination by proof padding)**.** *Let $\mathcal{L}$ be finite and $B$ decidable in polynomial time and closed under permutations of the parameters; assume* **??**. *With the constants $k_0$, $\beta$ of* **??**, *put $\bar{g}(m) := (g(m) + m + 1)(g(m) + m + 2)/2$, $\bar{g}'(m) := \beta \bar{g}(m) \log(\bar{g}(m) + 2)$ and $t^{+\#}(m) := m^{k_0} + m^{k_0}(t(m) + 1)$. For every hypothesis $p$ of $W_A^b[t, g]$ there is a hypothesis $p'$ whose axiom set is observational, with $|p'| \leq |p| + c$ and $p \mapsto p'$ injective, such that $p'$ is compatible with every observational $D$ with which $p$ is, under (b) with budget $2\bar{g}'(m) + m + 31$ and under (a1) with budget $\bar{g}'(m) + m + 16$, with membership time $t^{+\#}$. Hence, on observational data,*

$$W_A^b[t, g](D) \leq 2^c\, W_A^{\mathcal{L}_O, b}[t^{+\#}, 2\bar{g}' + m + 31](D), \qquad W_A^b[t, g](D) \leq 2^c\, W_A^{\mathcal{L}_O, a1}[t^{+\#}, \bar{g}' + m + 16](D),$$

*where $W_A^{\mathcal{L}_O, b}$ and $W_A^{\mathcal{L}_O, a1}$ are the sub-sums over hypotheses with observational axiom sets.*

*Proof.* By the second inequality of **??**(b) (Proposition E.21 of the earlier report), $p$ maps to the verifier $\mathcal{V}_p(\varphi, \gamma)$ that outputs acc if $\gamma$ codes a derivation of $\varphi$ from $B \cup A_p$ of material at most $g(|\varphi|)$, rej if it codes one of $\neg\varphi$, and $\bot$ otherwise; it runs in time $O(t^+)$, and every datum that $p$ derives has a certificate of length at most $\bar{g}'(|\varphi|)$ (this step uses **??**). Restrict $\mathcal{V}_p$ to observational $\varphi$, with output $\bot$ elsewhere (a linear-time syntax test). The third inequality maps a verifier $\mathcal{V}$ to $A_\mathcal{V} := \{\theta_\gamma \to \varphi : \mathcal{V}(\varphi, \gamma) = \text{acc}\} \cup \{\theta_\gamma \to \neg\varphi : \mathcal{V}(\varphi, \gamma) = \text{rej}\}$ (certificates of length at most $\bar{g}'$), a parameter-free set with membership time $t^{+\#}$. Each member is observational and logically equivalent to an observational theorem of $B \cup A_p$, so $B \cup A_\mathcal{V} \subseteq \mathrm{Cn}(A_p)$ is consistent. Each observational datum derived by $p$ within material $g$ has the derivation $\theta_\gamma$, $\theta_\gamma \to \varphi^b$, $\varphi^b$, of material $2|\gamma| + 29 + |\varphi^b|$ and largest instance $|\gamma| + 15 + |\varphi^b|$. Lengths add constants, and the maps are injective. □

Under (a1) we compare only fit classes, through Proposition A.11 below; a hypothesis-by-hypothesis translation is not claimed. Under (a2) and (b2), at a fixed budget, an observational substitute may need a table of the data below the size of its own members, so its description length need not be $|p| + O(1)$; again only fit classes are compared (Theorem A.14).

**The pairing structure.** $\mathfrak{U}$ is the least set that contains the words (the elements $t_w$ of $\mathfrak{T}$) and is closed under a formal pairing $(a, b) \mapsto \mathrm{pr}(a, b)$, where pr is injective and its values are new (not words). $\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1$ act on words as in $\mathfrak{T}$, and on pairs by $\mathtt{s}_0\,\mathrm{pr}(a, b) := a$ and $\mathtt{s}_1\,\mathrm{pr}(a, b) := \mathrm{pr}(b, a)$; $R$ is left free. In $\mathcal{L}_O$ let $\mathrm{Pair}(z) := \mathtt{s}_1\mathtt{s}_1 z = z$, $\mathrm{Std}(y) := \neg\mathrm{Pair}(y)$ and

$$\theta_2(a, b) := \forall y_\theta\, (\mathrm{Pair}(y_\theta) \to (\mathtt{s}_0 y_\theta = a \to (\mathtt{s}_0\mathtt{s}_1 y_\theta = b \to R(y_\theta)))),$$

where the bound variable $y_\theta$ occurs in no other formula below. It must be fresh: $\theta_2(a, b)$ is substituted for atoms inside formulas with their own quantified variables, and a bound variable named like one of them would capture it.

**Lemma A.10** (Pairing)**.** *In $\mathfrak{U}$, for every interpretation of $R$: (i)* Pair *holds exactly on the pairs, so* Std *holds exactly on the words, and $\mathfrak{U} \models \mathrm{Std}(\mathsf{e})$ and $\mathfrak{U} \models \forall y\,(\mathrm{Std}(y) \to \mathrm{Std}(\mathsf{s}_a y))$ for $a = 0, 1$; (ii) for all $a, b \in \mathfrak{U}$, $\mathrm{Pair}(z) \wedge \mathsf{s}_0 z = a \wedge \mathsf{s}_0\mathsf{s}_1 z = b$ holds iff $z = \mathrm{pr}(a, b)$; (iii) for all $a, b \in \mathfrak{U}$, $\theta_2(a, b)$ holds iff $R(\mathrm{pr}(a, b))$.*

*Proof.* (i) On words, $\mathsf{s}_1\mathsf{s}_1 t_v = t_{11v} \neq t_v$; on pairs, $\mathsf{s}_1\mathsf{s}_1\,\mathrm{pr}(a, b) = \mathsf{s}_1\,\mathrm{pr}(b, a) = \mathrm{pr}(a, b)$; and words are closed under $\mathsf{s}_0, \mathsf{s}_1$. (ii) A pair $z = \mathrm{pr}(c, d)$ has $\mathsf{s}_0 z = c$ and $\mathsf{s}_0\mathsf{s}_1 z = \mathsf{s}_0\,\mathrm{pr}(d, c) = d$; by injectivity of pr it equals $\mathrm{pr}(a, b)$ iff $(c, d) = (a, b)$. (iii) By (ii) the premise singles out $y_\theta = \mathrm{pr}(a, b)$. □

A computation on a fragment of $\mathfrak{U}$ with 800 elements (583200 triples), and independently on a fragment of 552 elements including pairs of pairs, finds no exception.

**Proposition A.11** (Ground pairing translation, reading (a1))**.** *Let $B = \emptyset$, and let $\mathcal{M}$ be an alternating machine with branching at most 2 that decides $X$ in space $s(m) \geq m$ and halts on every path, and let $A_{\mathcal{M}}$ be the ground axioms of Proposition E.23(b) of the earlier report in $\{\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1, R, \mathsf{C}, \mathsf{C}'\}$, where $\mathsf{C}(t_w, t_\varsigma)$ says that the configuration $\varsigma$ of $\mathcal{M}$ on $w$ accepts and $\mathsf{C}'(t_w, t_\varsigma)$ that it rejects. Let $A^O_{\mathcal{M}}$ replace every atom $\mathsf{C}(t_w, t_\varsigma)$ by $\theta_2(t_w, \mathsf{s}_0 t_\varsigma)$ and every $\mathsf{C}'(t_w, t_\varsigma)$ by $\theta_2(t_w, \mathsf{s}_1 t_\varsigma)$. Then $A^O_{\mathcal{M}}$ is observational and parameter-free with polynomial-time membership (by a decider of length $|\mathcal{M}| + c$); $A^O_{\mathcal{M}} \cup \Gamma_{f_X}$ is consistent; and every datum has a derivation from $A^O_{\mathcal{M}}$ by modus ponens alone with $2^{O(s(|w|)+1)}$ lines, each of size $O(s(|w|) + 1)$. Hence the fit class of the observational hypotheses of $W^{a1}_A$ at polynomial budgets is* EXPTIME*, as for all hypotheses.*

*Proof. Membership:* parse the fixed shape of the $\theta_2$-atoms, read off $t_w$, the tag letter and $t_\varsigma$, and run the check of Proposition E.23(b) there. *Derivations:* the derivations there use modus ponens only, between ground atoms and ground members; modus ponens is syntactic in the major premise, so translating atom by atom gives derivations from $A^O_{\mathcal{M}}$, and each atom grows by $O(1)$ symbols. *Consistency:* in $\mathfrak{U}$ put $R(t_w) := \mathbf{1}[w \in X]$, $R(\mathrm{pr}(t_w, t_{0\varsigma})) := \mathbf{1}[\varsigma \text{ accepts}]$, $R(\mathrm{pr}(t_w, t_{1\varsigma})) := \mathbf{1}[\varsigma \text{ rejects}]$, and $R$ false on all other pairs. By Lemma A.10(iii) each translated atom has the truth value of its original in the model of Proposition E.23(b) (the term algebra, with $\mathsf{C}, \mathsf{C}'$ the least fixed points), so $A^O_{\mathcal{M}}$ and $\Gamma_{f_X}$ are true in $\mathfrak{U}$. *Fit class:* $\supseteq$ EXPTIME by Corollary 6.17(b) there, with $A^O_{\mathcal{M}}$ in place of $A_{\mathcal{M}}$ and EXPTIME = APSPACE (Chandra et al., 1981); $\subseteq$ by Proposition E.23(a) there, which holds for every hypothesis satisfying (F1). □

A computation on an alternating machine for an interval game (positions are intervals of the word, the players alternately delete the first or last letter, and the existential player wins iff the last letter is 1) finds valid derivations, against an independent parsing decider, for all 2046 data with $1 \leq |w| \leq 10$, and all 14322 members with $|w| \leq 7$ recognised and true in $\mathfrak{U}$.

## A.4 One binary relation: two-stack programs stated backwards

For a word $u = u_1 \cdots u_k$ and a term $\tau$ write $u\tau$ for $\mathsf{s}_{u_1}(\cdots \mathsf{s}_{u_k}(\tau) \cdots)$; so $t_w = w\mathsf{e}$, and a pair of words is a pair of closed terms.

**Definition A.12** (Two-stack programs and their backward theories)**.** A *pattern* is $\pi = (r_\pi, \epsilon_r;\ l_\pi, \epsilon_l)$ with words $r_\pi, l_\pi$ and $\epsilon_r, \epsilon_l \in \{\mathrm{var}, \mathrm{exact}\}$. A pair of words $(r, l)$ *matches* $\pi$ with values $(Z_r, Z_l)$ if $r = r_\pi Z_r$ when $\epsilon_r = \mathrm{var}$, and $r = r_\pi$ (with $Z_r$ absent) when $\epsilon_r = \mathrm{exact}$, and likewise $l = l_\pi Z_l$ or $l = l_\pi$. A *rule* $\pi \Rightarrow \pi'$ is a pair of patterns with the same $\epsilon$'s; it is *linear* (each variable occurs once on each side), and applied to $(r_\pi Z_r, l_\pi Z_l)$ it gives $(r_{\pi'} Z_r, l_{\pi'} Z_l)$. A *two-stack program* $\mathcal{P}$ is a finite set of rules; it is *deterministic* if no pair of words matches the

current patterns of two rules. Programs carry a *tag* letter $\mathrm{tg} \in \{0,1\}$ at the front of $l$ (0 for an acceptance run, 1 for a rejection run), which every rule preserves, followed by a fixed-length *state code* $\mathrm{st}(q)$; the *halting prefixes* are $H_{\mathrm{acc}} := 0\,\mathrm{st}(\mathrm{ACC})$ and $H_{\mathrm{rej}} := 1\,\mathrm{st}(\mathrm{REJ})$, and the *start prefixes* are $\mathrm{tg}\,\mathrm{st}(\mathrm{start})$.

The *backward theory* of $\mathcal{P}$ in $\mathcal{L}_O \cup \{\mathsf{C}\}$, $\mathsf{C}$ a binary relation symbol, is

$$T_{\mathcal{P}}^{\mathsf{C}} := \{\forall Z_r \forall Z_l\,(\mathsf{C}(\pi') \to \mathsf{C}(\pi)) : (\pi \Rightarrow \pi') \in \mathcal{P}\} \cup \{\forall Z_r \forall Z_l\,\mathsf{C}(Z_r, H_{\mathrm{acc}} Z_l),\ \forall Z_r \forall Z_l\,\mathsf{C}(Z_r, H_{\mathrm{rej}} Z_l)\}$$
$$\cup\, \{\forall x \forall z\,(\mathsf{C}(x, 0\,\mathrm{st}(\mathrm{start})\,z) \to R(x)),\ \forall x \forall z\,(\mathsf{C}(x, 1\,\mathrm{st}(\mathrm{start})\,z) \to \neg R(x))\},$$

where $\mathsf{C}(\pi)$ is $\mathsf{C}(r_\pi Z_r, l_\pi Z_l)$ (without a variable where the pattern is exact) and the quantifiers range over the variables present. The two *output rules* carry a free certificate variable $z$; they are used only for programs with a certificate phase that consumes $z$ (Lemma A.13), never for the alternating programs in the proof of Theorem A.14(iv), which use exact output rules. The *observational translation* $T_{\mathcal{P}}^{O}$ replaces each atom $\mathsf{C}(a,b)$ by $\theta_2(a,b)$, prefixes the matrix of every member by $\mathrm{Std}(v) \to \cdots$ for each of its universally quantified variables $v$, and adds $\mathrm{Std}(\mathsf{e})$ and $\forall y\,(\mathrm{Std}(y) \to \mathrm{Std}(\mathsf{s}_a y))$ for $a = 0, 1$; it is a finite set of $\mathcal{L}_O$-sentences.

A forward theory (a start axiom, rules $\mathsf{C}(\pi) \to \mathsf{C}(\pi')$, an output rule reading the input from a component) must keep the input word in the configuration until the end, which needs a relation of higher arity. Stated backwards, $\mathsf{C}$ means "this configuration leads to a halting configuration of the right kind", and the input is read only by the output rules, at the start configuration.

**Lemma A.13** (Every verifier is a deterministic two-stack program)**.** *Let $\mathcal{V}$ be a verifier: a deterministic multitape machine that halts on every pair $(x, z)$ of words with output acc, rej or $\perp$, such that no $x$ has both an accepting and a rejecting certificate ($f_{\mathcal{V}}$ is consistent); let $X_{\mathrm{acc}} := \{x : \exists z\ \mathcal{V}(x,z) = \mathrm{acc}\}$ and $X_{\mathrm{rej}}$ likewise. There is a deterministic two-stack program $\mathcal{P}_{\mathcal{V}}$ such that for all words $x, z$ and both tags: (i) every rule is linear, the current patterns are pairwise disjoint, and no rule applies to a pair whose $l$ begins with a halting prefix or with the code of the state $\perp$; (ii) the run from the* start pair *$(x, \mathrm{tg}\,\mathrm{st}(\mathrm{start})\,z)$ reaches a pair with prefix $H_{\mathrm{acc}}$ (for $\mathrm{tg} = 0$) iff $\mathcal{V}(x,z) = \mathrm{acc}$, and one with prefix $H_{\mathrm{rej}}$ (for $\mathrm{tg} = 1$) iff $\mathcal{V}(x,z) = \mathrm{rej}$; (iii) the run has $N_{\mathrm{run}} = O(T_1 + |x| + |z|)$ steps, every pair on it has length $O(S_1 + |x| + |z|)$, and each step changes $O(1)$ letters, where $T_1$ and $S_1$ are the time and space of a one-tape simulation of $\mathcal{V}$ (so $T_1$ is polynomial in the time of $\mathcal{V}$ and $S_1 = O(S_{\mathcal{V}} + |x| + |z|)$).*

*Proof. (1) The one-tape machine.* By the one-tape simulation of multitape machines (standard; cf. Hartmanis and Stearns, 1965) there is a deterministic one-tape machine $\mathcal{M}_1$ that, started on the tape $\mathtt{Z}_{z_m} \cdots \mathtt{Z}_{z_1} \# x_1 \cdots x_{|x|}$ (the certificate reversed, in letters $\mathtt{Z}_0, \mathtt{Z}_1$, then a separator, then the input) with the head on cell 0, halts in state ACC, REJ or $\perp$ according to $\mathcal{V}(x,z)$; rearranging the input format costs $O((|x| + |z|)^2)$ steps, and the simulation is polynomial in the time of $\mathcal{V}$ and linear in its space. Bring $\mathcal{M}_1$ to a normal form: its tape alphabet $\Sigma_{\mathcal{M}}$ (without the blank $\square$) contains $\#, 0, 1, \mathtt{Z}_0, \mathtt{Z}_1$ and a symbol $\mathtt{B}$ on which $\mathcal{M}_1$ has exactly the transitions it has on $\square$, and wherever $\mathcal{M}_1$ would write $\square$ it writes $\mathtt{B}$; so every transition writes a symbol of $\Sigma_{\mathcal{M}}$. Moving left on cell 0 leaves the head there. Then on every run from a start tape the non-blank cells form an initial segment $[0, \mathrm{nb})$ and the head position hp satisfies $\mathrm{hp} \le \mathrm{nb}$: a step at $\mathrm{hp} < \mathrm{nb}$ keeps both properties, and a step at $\mathrm{hp} = \mathrm{nb}$ writes a symbol of $\Sigma_{\mathcal{M}}$ there, so nb grows by one.

*(2) Coding.* Fix injective codes $\mathrm{st} : Q \to \{0,1\}^{k_Q}$ (the states of $\mathcal{M}_1$ and the phase states $\mathrm{start}, \mathrm{cert}, \mathrm{sep}, \mathrm{inp}, \mathrm{rew}$) and $\mathrm{blk} : \Sigma_{\mathcal{M}} \to \{0,1\}^{k_B}$. A configuration of $\mathcal{M}_1$ with state $q$, tape $a_0 \cdots a_{\mathrm{nb}-1} \square\square \cdots$ and head $\mathrm{hp} \le \mathrm{nb}$ is coded, with tag tg, as

$$[\![q, a, \mathrm{hp}]\!]_{\mathrm{tg}} := \big(\mathrm{blk}(a_{\mathrm{hp}}) \cdots \mathrm{blk}(a_{\mathrm{nb}-1}),\quad \mathrm{tg}\,\mathrm{st}(q)\,\mathrm{blk}(a_{\mathrm{hp}-1}) \cdots \mathrm{blk}(a_0)\big) :$$

$r$ holds the cells from the head on, $l$ the tag, the state and the cells to the left of the head, nearest first; $r = \varepsilon$ means that the head reads $\square$.

*(3) Rules*, for each tag, with $\mathrm{cd}(q) := \mathrm{tg}\,\mathrm{st}(q)$; a pattern is written $(\rho; \lambda)$, and "$\varepsilon \,|\, \text{exact}$" or "$\mathrm{cd}(q) \,|\, \text{exact}$" marks an exact component:

- *start:* $(Z_r;\, \mathrm{cd}(\mathrm{start})Z_l) \Rightarrow (\mathrm{blk}(\#)Z_r;\, \mathrm{cd}(\mathrm{cert})Z_l)$, which pushes the separator onto $r$, above the raw input;
- *phase cert:* $(Z_r;\, \mathrm{cd}(\mathrm{cert})aZ_l) \Rightarrow (\mathrm{blk}(\mathtt{Z}_a)Z_r;\, \mathrm{cd}(\mathrm{cert})Z_l)$ for $a \in \{0,1\}$ (a raw certificate letter below the state code becomes a block on $r$), and $(Z_r;\, \mathrm{cd}(\mathrm{cert}) \,|\, \text{exact}) \Rightarrow (Z_r;\, \mathrm{cd}(\mathrm{sep}) \,|\, \text{exact})$;
- *phase sep:* $(\mathrm{blk}(\mathtt{Z}_a)Z_r;\, \mathrm{cd}(\mathrm{sep})Z_l) \Rightarrow (Z_r;\, \mathrm{cd}(\mathrm{sep})\mathrm{blk}(\mathtt{Z}_a)Z_l)$, and $(\mathrm{blk}(\#)Z_r;\, \mathrm{cd}(\mathrm{sep})Z_l) \Rightarrow (Z_r;\, \mathrm{cd}(\mathrm{inp})\mathrm{blk}(\#)Z_l)$;
- *phase inp:* $(aZ_r;\, \mathrm{cd}(\mathrm{inp})Z_l) \Rightarrow (Z_r;\, \mathrm{cd}(\mathrm{inp})\mathrm{blk}(a)Z_l)$ for $a \in \{0,1\}$, and $(\varepsilon \,|\, \text{exact};\, \mathrm{cd}(\mathrm{inp})Z_l) \Rightarrow (\varepsilon \,|\, \text{exact};\, \mathrm{cd}(\mathrm{rew})Z_l)$;
- *phase rew:* $(Z_r;\, \mathrm{cd}(\mathrm{rew})\mathrm{blk}(e)Z_l) \Rightarrow (\mathrm{blk}(e)Z_r;\, \mathrm{cd}(\mathrm{rew})Z_l)$ for $e \in \{\mathtt{Z}_0, \mathtt{Z}_1, \#, 0, 1\}$, and $(Z_r;\, \mathrm{cd}(\mathrm{rew}) \,|\, \text{exact}) \Rightarrow (Z_r;\, \mathrm{cd}(q_0) \,|\, \text{exact})$;
- *the machine:* for each transition $\delta(q,a) = (q', b, \mathrm{dir})$, with read pattern $\rho := \mathrm{blk}(a)Z_r$ if $a \in \Sigma_{\mathcal{M}}$ and $\rho := \varepsilon \,|\, \text{exact}$ if $a = \square$, and $\rho'$ the remainder ($Z_r$, respectively $\varepsilon \,|\, \text{exact}$): for $\mathrm{dir} = R$, $(\rho;\, \mathrm{cd}(q)Z_l) \Rightarrow (\rho';\, \mathrm{cd}(q')\mathrm{blk}(b)Z_l)$; for $\mathrm{dir} = S$, $(\rho;\, \mathrm{cd}(q)Z_l) \Rightarrow (\mathrm{blk}(b)\rho';\, \mathrm{cd}(q')Z_l)$; for $\mathrm{dir} = L$, for each $e \in \Sigma_{\mathcal{M}}$, $(\rho;\, \mathrm{cd}(q)\mathrm{blk}(e)Z_l) \Rightarrow (\mathrm{blk}(e)\mathrm{blk}(b)\rho';\, \mathrm{cd}(q')Z_l)$, and at the left end $(\rho;\, \mathrm{cd}(q) \,|\, \text{exact}) \Rightarrow (\mathrm{blk}(b)\rho';\, \mathrm{cd}(q') \,|\, \text{exact})$.

(i) Linearity is visible in the list. Rules with different tags or state codes differ in a fixed-length prefix of $l$. Within one state, the read patterns $\mathrm{blk}(a)Z_r$ (fixed length, injective code) and $\varepsilon \,|\, \text{exact}$ are pairwise disjoint, as are, within one left-moving transition, the neighbour patterns $\mathrm{blk}(e)Z_l$ and the exact left-end pattern; in phase cert the raw-letter patterns $\mathrm{cd}(\mathrm{cert})aZ_l$ and the exact $\mathrm{cd}(\mathrm{cert})$ are disjoint, and in phase inp the raw-letter patterns $aZ_r$ and the exact $\varepsilon$. There are no rules for ACC, REJ and $\bot$, and fixed-length state codes prevent a spurious halting prefix. (ii) *Phases.* From $(x, \mathrm{cd}(\mathrm{start})z)$ the start rule gives $(\mathrm{blk}(\#)x, \mathrm{cd}(\mathrm{cert})z)$; phase cert moves the letters of $z$, first letter first, onto $r$, ending in $(\mathrm{blk}(\mathtt{Z}_{z_m}) \cdots \mathrm{blk}(\mathtt{Z}_{z_1})\mathrm{blk}(\#)x,\ \mathrm{cd}(\mathrm{sep}))$; phase sep moves these blocks to $l$, the separator last, ending in $(x,\ \mathrm{cd}(\mathrm{inp})\mathrm{blk}(\#)\mathrm{blk}(\mathtt{Z}_{z_1}) \cdots \mathrm{blk}(\mathtt{Z}_{z_m}))$; phase inp converts the raw letters of $x$; and phase rew moves all blocks back to $r$ and ends exactly at the left end in $[\![q_0,\ \mathtt{Z}_{z_m} \cdots \mathtt{Z}_{z_1}\#x,\ 0]\!]_{\mathrm{tg}}$, the code of the start configuration of $\mathcal{M}_1$. *Steps.* For a configuration with $\mathrm{hp} \le \mathrm{nb}$ and a non-blank initial segment, exactly one rule applies to its code, namely the rule of the transition taken, with the neighbour case given by the cell $\mathrm{hp} - 1$ or the left end, and its result is the code of the successor: for $\mathrm{dir} = R$ the written $b$ moves to the top of $l$; for $S$ it replaces the top of $r$; for $L$ the left neighbour $e$ moves from $l$ to $r$ above $b$; and at $a = \square$ ($r = \varepsilon$) the written $b$ becomes the new last cell. Reading $\mathtt{B}$ is reading the block $\mathrm{blk}(\mathtt{B})$, with the transitions of $\square$. By (1) the invariant is preserved. So the run of $\mathcal{P}_{\mathcal{V}}$ from the start pair passes through the codes of the run of $\mathcal{M}_1$ and stops exactly when $\mathcal{M}_1$ halts. The tag is preserved, so a 0-tagged run can reach $H_{\mathrm{acc}}$ only through ACC, and a 1-tagged run $H_{\mathrm{rej}}$ only through REJ. (iii) The phases take $2|z| + |x| + (|z| + |x| + 1) + O(1)$ steps, the machine $T_1$; every rule changes $O(k_Q + k_B)$ letters. $\square$

**Theorem A.14** (One binary theoretical relation suffices under charged deduction)**.** *Let $B = \emptyset$, let $\mathcal{V}$ be a verifier as in Lemma A.13, $\mathcal{P} := \mathcal{P}_{\mathcal{V}}$, and let $\mathsf{C}^{\mathfrak{T}} \subseteq \mathfrak{T}^2$ be the least set of pairs of words that contains all pairs with $l$-prefix $H_{\mathrm{acc}}$ or $H_{\mathrm{rej}}$ and is closed under the rules read backwards (if the result of a rule instance is in the set, so is its current pair). Write $T^{\mathsf{C}}_{\mathcal{V}} := T^{\mathsf{C}}_{\mathcal{P}}$ and $T^{O}_{\mathcal{V}} := T^{O}_{\mathcal{P}}$.*

*(i)* Relation version. $(\mathfrak{T}, \mathsf{C}^{\mathfrak{T}}, R := X_{\mathrm{acc}}) \models T_{\mathcal{V}}^{\mathsf{C}} \cup \Gamma_{f_{\mathcal{V}}}$ *and* $(\mathfrak{T}, \mathsf{C}^{\mathfrak{T}}, R := \{w : w \notin X_{\mathrm{rej}}\}) \models T_{\mathcal{V}}^{\mathsf{C}}$. *So* $T_{\mathcal{V}}^{\mathsf{C}}$ *is a finite theory in* $\mathcal{L}_O \cup \{\mathsf{C}\}$ *about the named objects, consistent with* $\Gamma_{f_{\mathcal{V}}}$, *and* $T_{\mathcal{V}}^{\mathsf{C}} \vdash R(t_w)$ *iff* $w \in X_{\mathrm{acc}}$, $T_{\mathcal{V}}^{\mathsf{C}} \vdash \neg R(t_w)$ *iff* $w \in X_{\mathrm{rej}}$.

*(ii)* Observational version. *In* $\mathfrak{U}$ *put* $R(t_w) := \mathbf{1}[w \in X_{\mathrm{acc}}]$, $R(\mathrm{pr}(a,b)) := \mathbf{1}[(a,b) \in \mathsf{C}^{\mathfrak{T}}]$ *for words* $a, b$, *and* $R$ *false on all other pairs. Then* $\mathfrak{U} \models T_{\mathcal{V}}^{O} \cup \Gamma_{f_{\mathcal{V}}}$, *and with* $R(t_w) := \mathbf{1}[w \notin X_{\mathrm{rej}}]$ *instead,* $\mathfrak{U} \models T_{\mathcal{V}}^{O}$. *So* $T_{\mathcal{V}}^{O}$ *is a finite set of* $\mathcal{L}_O$*-sentences, consistent with* $\Gamma_{f_{\mathcal{V}}}$, *proving exactly the literals* $\varphi_w$ *with* $w \in X_{\mathrm{acc}}$ *and* $\neg\varphi_w$ *with* $w \in X_{\mathrm{rej}}$.

*(iii)* Derivations. *If* $\mathcal{V}(w,z) \in \{\mathrm{acc}, \mathrm{rej}\}$, *the corresponding literal has a* $\mathsf{K}$*-derivation from* $T_{\mathcal{V}}^{\mathsf{C}}$ *(respectively* $T_{\mathcal{V}}^{O}$*) by instances of the quantifier axiom A4 and modus ponens only, with* $O(N_{\mathrm{run}})$ *(respectively* $O(N_{\mathrm{run}} + \ell)$*) lines,* $N_{\mathrm{run}}$ *the number of steps of Lemma A.13(iii) and* $\ell$ *the largest length of a pair on the run, each line of size* $O(\ell + 1)$*; its nonlogical instances are members of the theory, each of size at most* $\mathrm{mx}_{\mathcal{V}}$ *and together at most* $\mathrm{tot}_{\mathcal{V}}$.

*(iv)* Fit classes. *Hence, on literal sequences, finite theories of either kind are compatible with all of* $D^X$ *for every decidable* $X$ *under (a2) and (b2), at every nondecreasing unbounded budget with* $h(m) \geq m+1$ *(with a finite table of the data below the budget threshold added as literals; with a budget of at least* $\mathrm{mx}_{\mathcal{V}}$*, respectively* $\mathrm{tot}_{\mathcal{V}}$*, everywhere no table is needed); and, at polynomial budgets, for every* $X \in \mathsf{NP} \cap \mathsf{coNP}$ *under (b) (material* $O(N_{\mathrm{run}}\ell)$*, polynomial for polynomial-time verifiers), the fit class of all hypotheses (**??**). Under (a1): for every* $X \in \mathsf{EXPTIME}$ *there are a finite theory in* $\mathcal{L}_O \cup \{\mathsf{C}\}$ *about the named objects and a finite set of* $\mathcal{L}_O$*-sentences, each consistent with* $\Gamma_{f_X}$, *from which every datum* $\varphi_w^{\mathbf{1}[w \in X]}$ *has a* $\mathsf{K}$*-derivation whose every line, logical axiom instances included, has size* $O(s(|w|) + |w|)$ *for a polynomial* $s$*; so at polynomial budgets the (a1) fit class of both kinds of finite theory is* $\mathsf{EXPTIME}$*, the class of all hypotheses. Membership is linear-time (finite sets of sentences) and* **??** *holds (sentences).*

*Proof.* (i) The rules hold in $(\mathfrak{T}, \mathsf{C}^{\mathfrak{T}})$ by the closure of $\mathsf{C}^{\mathfrak{T}}$, the halting axioms by its base. For the output rules we show: *a start pair* $\sigma = (x, \mathrm{tg}\,\mathrm{st}(\mathrm{start})\,z)$, $z$ *any word, lies in* $\mathsf{C}^{\mathfrak{T}}$ *iff its run reaches a pair with prefix* $H_{\mathrm{acc}}$ *(*$\mathrm{tg} = 0$*) or* $H_{\mathrm{rej}}$ *(*$\mathrm{tg} = 1$*).* Let $S$ be the set of pairs whose deterministic run reaches a halting prefix. $S$ contains the base and is closed under the backward rules: if the result $\pi'(Z_r, Z_l)$ of a rule instance is in $S$, then $\pi(Z_r, Z_l)$ matches that rule and, by disjointness (Lemma A.13(i), on all pairs of words), no other, so its run moves to $\pi'(Z_r, Z_l)$ and continues into the base; so $\mathsf{C}^{\mathfrak{T}} \subseteq S$. Conversely $S \subseteq \mathsf{C}^{\mathfrak{T}}$ by induction on the number of steps to the base. With Lemma A.13(ii): $(x, 0\,\mathrm{st}(\mathrm{start})z) \in \mathsf{C}^{\mathfrak{T}}$ iff $\mathcal{V}(x,z) = \mathrm{acc}$, which implies $x \in X_{\mathrm{acc}}$, so the first output rule holds with $R = X_{\mathrm{acc}}$; and $(x, 1\,\mathrm{st}(\mathrm{start})z) \in \mathsf{C}^{\mathfrak{T}}$ iff $\mathcal{V}(x,z) = \mathrm{rej}$, which implies $x \in X_{\mathrm{rej}}$, hence $x \notin X_{\mathrm{acc}}$ by the consistency of $f_{\mathcal{V}}$, so the second holds. $\Gamma_{f_{\mathcal{V}}}$ holds since $X_{\mathrm{acc}} \cap X_{\mathrm{rej}} = \emptyset$. The second model is checked in the same way. The two models give that $T_{\mathcal{V}}^{\mathsf{C}} \vdash R(t_w)$ only if $w \in X_{\mathrm{acc}}$ and $T_{\mathcal{V}}^{\mathsf{C}} \vdash \neg R(t_w)$ only if $w \in X_{\mathrm{rej}}$; the converses are (iii).

(ii) For values of its variables in $\mathrm{Std}^{\mathfrak{U}} = \mathfrak{T}$ (Lemma A.10(i)), each member of $T_{\mathcal{V}}^{O}$ says what the corresponding member of $T_{\mathcal{V}}^{\mathsf{C}}$ says with $\mathsf{C}(a,b)$ read as $R(\mathrm{pr}(a,b))$ (Lemma A.10(iii)); terms built from words by $\mathtt{s}_0, \mathtt{s}_1$ stay words and are evaluated as in $\mathfrak{T}$. For values outside $\mathfrak{T}$ a Std premise fails. The Std axioms hold by Lemma A.10(i).

(iii) Let $\sigma_0, \ldots, \sigma_{N_{\mathrm{run}}}$ be the run from $\sigma_0 = (t_w, \mathrm{tg}\,\mathrm{st}(\mathrm{start})\,t_z)$. Instantiate the halting axiom at $\sigma_{N_{\mathrm{run}}}$ (two A4 instances and two modus ponens steps) to get $\mathsf{C}(\sigma_{N_{\mathrm{run}}})$; for $i = N_{\mathrm{run}} - 1, \ldots, 0$, instantiate the rule that applies to $\sigma_i$ at its values (one A4 instance and one modus ponens step per variable) and apply modus ponens to $\mathsf{C}(\sigma_{i+1})$, getting $\mathsf{C}(\sigma_i)$; finally instantiate the output rule at $x := t_w$, $z := t_z$ and apply modus ponens. In $T_{\mathcal{V}}^{O}$ the instantiated members carry Std premises

for the values, all words: derive $\mathrm{Std}(t_v)$ for every suffix $v$ of every component from $\mathrm{Std}(\mathsf{e})$ and the propagation axioms (one A4 instance and two modus ponens steps per letter, keeping the set of words with a derived Std fact closed under suffixes, each step adding $O(1)$ letters), and discharge them by modus ponens. Every line is a member, an instance of a member at the values of one pair on the run, or an atom about such a pair, so it has size $O(\ell+1)$.

(iv) For a decidable $X$ take $\mathcal{V}(x,z) :=$ the decision of $X$ on $x$ (ignoring $z$). Under (a2) and (b2) every nonlogical instance is a member of the finite theory, so the budget needed is a constant; data below it are listed as table literals, of size $|\varphi_w|$ or $|\varphi_w|+1$, which keeps the theory finite and true in the models above. Under (b) the material is at most the number of distinct instances, $O(N_{\mathrm{run}})$, times their size, $O(\ell)$; for $X \in \mathsf{NP} \cap \mathsf{coNP}$ combine polynomial-time verifiers for $X$ and its complement into one consistent verifier, and the upper bound is that of all hypotheses (**??**).

*Reading (a1). Step 1 (the machine).* An alternating one-tape machine has states partitioned into existential and universal states and the halting states ACC and REJ, and a transition relation giving, for each non-halting state $q$ and symbol $a \in \Sigma_{\mathcal{M}} \cup \{\Box\}$, a list $\delta(q,a)$ of one or two moves $(q_i, b_i, \mathrm{dir}_i)$. A configuration *accepts* if its state is ACC, or it is existential and some successor accepts, or it is universal and all successors accept (least fixed point); when every computation path halts, the computation tree from a configuration is finite (König's lemma), acceptance is decided by induction on it, and a configuration that does not accept *rejects*. Normal form: (1) as in Lemma A.13(1) ($\mathsf{B}$ has the transitions of $\Box$, only symbols of $\Sigma_{\mathcal{M}}$ are written, a left move at cell 0 stays), so $\mathrm{hp} \le \mathrm{nb}$ on every path from a start tape; (2) $\delta(q,a)$ is nonempty for every non-halting $q$ and every $a$ (add a move to REJ at an existential state, and to ACC at a universal state, where none exists; this does not change acceptance); (3) every computation path from a start configuration (tape $\#x$, head on cell 0, state $q_0$) halts. Such a machine exists for every $X \in \mathsf{EXPTIME}$. By $\mathsf{EXPTIME} = \mathsf{APSPACE}$ (Chandra et al., 1981), $X$ is decided by a multitape alternating machine $\mathcal{M}$ in polynomial space $S(m)$. *Clock.* $\mathcal{M}$ has at most $N_{\mathrm{cf}}(m) = 2^{c_{\mathcal{M}}(S(m)+\log m)}$ configurations on inputs of length $m$. Let $A_j$ be the set of configurations with an accepting computation tree of height at most $j$; $A_j \subseteq A_{j+1}$, the accepting configurations are $\bigcup_j A_j$, and the sequence stabilises after at most $N_{\mathrm{cf}}(m)$ strict increases, so every accepting configuration has an accepting tree of height at most $N_{\mathrm{cf}}(m)$. Add a binary step counter of $c_{\mathcal{M}}(S(m)+\log m)+1$ bits on an extra tape, and go to REJ when it overflows. If $\mathcal{M}$ accepts $x$, the clocked machine follows an accepting tree of height at most $N_{\mathrm{cf}}(m)$ and accepts; if the clocked machine accepts, its accepting tree contains no overflow (overflow leads to REJ, which an accepting tree does not contain) and is an accepting tree of $\mathcal{M}$. Every path of the clocked machine halts. *One tape.* Simulate the tapes on tracks of one tape, each step of the clocked machine by a deterministic sequence of steps (existential states with one successor) that ends with the branching step; the computation trees correspond, so acceptance is preserved and all paths halt, in space $O(S+\log m)$. The normal form (1)–(2) is a local change. Call the result $\mathcal{M}_1$, and split moves so that $|\delta(q,a)| \le 2$.

*Step 2 (the compile).* Fix codes st (states of $\mathcal{M}_1$ and the phase states $\mathrm{start}, \mathrm{inp}, \mathrm{rew}$) and blk as in Lemma A.13(2). For each tag, with $\mathrm{cd}(q) := \mathrm{tg}\,\mathrm{st}(q)$:

- *start:* $(Z_r;\, \mathrm{cd}(\mathrm{start})Z_l) \Rightarrow (Z_r;\, \mathrm{cd}(\mathrm{inp})\,\mathrm{blk}(\#)Z_l)$: the separator goes onto $l$, and there is no certificate phase;
- *phase inp:* $(aZ_r;\, \mathrm{cd}(\mathrm{inp})Z_l) \Rightarrow (Z_r;\, \mathrm{cd}(\mathrm{inp})\mathrm{blk}(a)Z_l)$ for $a \in \{0,1\}$, and $(\varepsilon \,|\, \mathrm{exact};\, \mathrm{cd}(\mathrm{inp})Z_l) \Rightarrow (\varepsilon \,|\, \mathrm{exact};\, \mathrm{cd}(\mathrm{rew})Z_l)$;
- *phase rew:* $(Z_r;\, \mathrm{cd}(\mathrm{rew})\mathrm{blk}(e)Z_l) \Rightarrow (\mathrm{blk}(e)Z_r;\, \mathrm{cd}(\mathrm{rew})Z_l)$ for $e \in \{\#,0,1\}$, and $(Z_r;\, \mathrm{cd}(\mathrm{rew}) \,|\, \mathrm{exact}) \Rightarrow (Z_r;\, \mathrm{cd}(q_0) \,|\, \mathrm{exact})$;

- *the machine:* for each non-halting $q$ and $a \in \Sigma_{\mathcal{M}} \cup \{\Box\}$, with read pattern $\rho$ and remainder $\rho'$ as in Lemma A.13(3), the *neighbour cases* are $\nu \in \{\mathrm{blk}(e)Z_l : e \in \Sigma_{\mathcal{M}}\} \cup \{| \,\mathrm{exact}\}$ if some move in $\delta(q,a)$ goes left, and the single case $\nu = Z_l$ otherwise. For each case the current pattern is $\pi := (\rho;\, \mathrm{cd}(q)\nu)$, and each move $(q_i, b_i, \mathrm{dir}_i)$ gives a next pattern $\pi_i'$: $(\rho';\, \mathrm{cd}(q_i)\mathrm{blk}(b_i)\nu)$ for $R$; $(\mathrm{blk}(b_i)\rho';\, \mathrm{cd}(q_i)\nu)$ for $S$; $(\mathrm{blk}(e)\mathrm{blk}(b_i)\rho';\, \mathrm{cd}(q_i)Z_l)$ for $L$ when $\nu = \mathrm{blk}(e)Z_l$; and $(\mathrm{blk}(b_i)\rho';\, \mathrm{cd}(q_i)\,|\,\mathrm{exact})$ for $L$ at the left end. The rule is *conjunctive*, $\pi \Rightarrow \{\pi_1', \pi_2'\}$ with backward form $\forall Z_r \forall Z_l\, (\mathsf{C}(\pi_1') \to (\mathsf{C}(\pi_2') \to \mathsf{C}(\pi)))$, when all successors are needed: at universal states for tag 0 and at existential states for tag 1. Otherwise there is one rule $\pi \Rightarrow \pi_i'$ per move, with backward form $\forall Z_r \forall Z_l\, (\mathsf{C}(\pi_i') \to \mathsf{C}(\pi))$; a single move gives a single rule;
- *halting axioms:* $\forall Z_r \forall Z_l\, \mathsf{C}(Z_r, H_{\mathrm{acc}} Z_l)$ and $\forall Z_r \forall Z_l\, \mathsf{C}(Z_r, H_{\mathrm{rej}} Z_l)$;
- *exact output rules:* $\forall x\, (\mathsf{C}(x,\, 0\,\mathrm{st}(\mathrm{start})) \to R(x))$ and $\forall x\, (\mathsf{C}(x,\, 1\,\mathrm{st}(\mathrm{start})) \to \neg R(x))$, where $\mathrm{tg\,st}(\mathrm{start})$ stands for the closed term $t_{\mathrm{tg\,st}(\mathrm{start})}$. In the observational version they are $\forall x\, (\mathrm{Std}(x) \to (\theta_2(x, t_{0\,\mathrm{st}(\mathrm{start})}) \to R(x)))$ and its rejecting twin.

Call the result $\mathcal{A}$, $T_{\mathcal{A}}^{\mathsf{C}}$ its backward theory (rules, halting axioms, exact output rules) and $T_{\mathcal{A}}^{O}$ its observational translation (Definition A.12). The current patterns of distinct cases (state, read symbol, neighbour case, tag) are pairwise disjoint, by the argument of Lemma A.13(i); the two rules of a non-conjunctive branching share one current pattern.

*Step 3 (the meaning of* $\mathsf{C}$ *at reachable pairs).* Let $\mathsf{C}^{\mathfrak{T}}$ be the least set of pairs of words that contains the pairs with a halting prefix and is closed under the backward rules (a conjunctive rule adds $\pi$ when all its next pairs are in, the others when their one next pair is in). Let Reach be the set of pairs reachable by the rules from the *exact* start pairs $(x,\, \mathrm{tg\,st}(\mathrm{start}))$, $x$ a word. Every pair in Reach is a phase pair of the run from a start pair, or the code $[\![q, a, \mathrm{hp}]\!]_{\mathrm{tg}}$ of a configuration of $\mathcal{M}_1$ reachable from $\#x$: the start rule gives $(x,\, \mathrm{cd}(\mathrm{inp})\mathrm{blk}(\#))$, phase inp gives $(\varepsilon,\, \mathrm{cd}(\mathrm{rew})\mathrm{blk}(x_{|x|}) \cdots \mathrm{blk}(x_1)\mathrm{blk}(\#))$, phase rew gives $(\mathrm{blk}(\#)\mathrm{blk}(x_1) \cdots \mathrm{blk}(x_{|x|}),\, \mathrm{cd}(q_0)) = [\![q_0, \#x, 0]\!]_{\mathrm{tg}}$, and from a configuration code the patterns that match are exactly those of its case, whose next pairs are the codes of its successors (the step argument of Lemma A.13). For $\sigma \in \mathrm{Reach}$ let $\mathrm{acc}(\sigma)$ mean that the configuration of $\sigma$ accepts (for a phase pair: the start configuration it leads to accepts). *Claim: for* $\sigma \in \mathrm{Reach}$ *with tag* 0, $\sigma \in \mathsf{C}^{\mathfrak{T}}$ *iff* $\mathrm{acc}(\sigma)$*; with tag* 1, $\sigma \in \mathsf{C}^{\mathfrak{T}}$ *iff not* $\mathrm{acc}(\sigma)$. ($\Leftarrow$, tag 0.) By induction on the height of the finite computation tree of $\sigma$, phase steps counting as deterministic steps: a halting accepting configuration has prefix $H_{\mathrm{acc}}$, so it is in the base; at an existential configuration some successor accepts and the rule for that move puts $\sigma$ in; at a universal one all successors accept and the conjunctive rule puts $\sigma$ in; a deterministic or phase step uses its single rule. ($\Rightarrow$, tag 0.) The set $S_0 := \{\sigma \in \mathrm{Reach} : \text{tag } 0,\ \mathrm{acc}(\sigma)\} \cup \{\text{0-tagged pairs outside Reach}\} \cup \{\text{all 1-tagged pairs}\}$ contains the base (a reachable pair with prefix $H_{\mathrm{acc}}$ is a configuration in state ACC) and is closed under the backward rules. Take a rule instance with current pair $\sigma$ whose next pairs (all of them for a conjunctive rule, its one next pair otherwise) lie in $S_0$. If $\sigma$ is 1-tagged or outside Reach, $\sigma \in S_0$. Otherwise $\sigma \in \mathrm{Reach}$ is 0-tagged, the rule belongs to the case of $\sigma$, and its next pairs are codes of successors of $\sigma$, hence in Reach and 0-tagged, hence accepting. A conjunctive rule at tag 0 sits at a universal or deterministic configuration, all of whose successors are then accepting; a non-conjunctive one sits at an existential configuration with an accepting successor. Either way $\mathrm{acc}(\sigma)$, so $\mathsf{C}^{\mathfrak{T}} \subseteq S_0$. (Tag 1.) The same, with acceptance and rejection, existential and universal exchanged: at tag 1 the conjunctive rules sit at existential states. So for the exact start pairs, $(x, 0\,\mathrm{st}(\mathrm{start})) \in \mathsf{C}^{\mathfrak{T}}$ iff $\mathcal{M}_1$ accepts $\#x$ iff $x \in X$, and $(x, 1\,\mathrm{st}(\mathrm{start})) \in \mathsf{C}^{\mathfrak{T}}$ iff $x \notin X$.

*Step 4 (theories, truth, derivations).* By Step 3, $(\mathfrak{T}, \mathsf{C}^{\mathfrak{T}}, R := X) \models T_{\mathcal{A}}^{\mathsf{C}} \cup \Gamma_{f_X}$: the rules by closure, the halting axioms by the base, the exact output rules by the Claim. So $T_{\mathcal{A}}^{\mathsf{C}}$ is about the

named objects, and it proves $\varphi^b_w$ only if $b = \mathbf{1}[w \in X]$. In $\mathfrak{U}$ put $R(t_w) := \mathbf{1}[w \in X]$, $R(\mathrm{pr}(a,b)) := \mathbf{1}[(a,b) \in \mathsf{C}^{\mathfrak{T}}]$ for words $a, b$, and $R$ false on other pairs; as in (ii), $\mathfrak{U} \models T^O_{\mathcal{A}} \cup \Gamma_{f_X}$ (the ground term $t_{\mathrm{tg\,st(start)}}$ of the exact output rule is a word, so $\theta_2(x, t_{\mathrm{tg\,st(start)}})$ holds iff $(x, \mathrm{tg\,st(start)}) \in \mathsf{C}^{\mathfrak{T}}$). For $w \in X$ fix an accepting computation tree of $\mathcal{M}_1$ on $\#w$ (all successors at universal nodes, one accepting successor at existential nodes) and the phase pairs before it; derive $\mathsf{C}(\sigma)$ for its nodes bottom-up, a leaf by the halting axiom (two A4 instances, two modus ponens steps), an inner node by the instance of its rule at its values and one modus ponens step per premise; then apply the exact output rule at $x := t_w$ and modus ponens. Rejection is the same with a rejecting tree and tag 1. In $T^O_{\mathcal{A}}$ the Std premises are discharged as in (iii). The number of lines may be exponential, but every line is a member, an instance of a member at the values of one reachable pair, or an atom about one, so every line, logical instances included, has size $O(S_1 + |w|)$, a polynomial. That is reading (a1) at a polynomial budget; the upper bound $\subseteq$ EXPTIME is **??**. □

**Why the output rules must be exact for alternating programs.** Without a certificate phase nothing consumes the variable $z$ of the output rules of Definition A.12. Phase rew rewinds every block-aligned $z$ onto the tape, so $\mathcal{M}_1$ runs on $z'\#x$, a tape about which the normal form says nothing. If $\mathcal{M}_1$ has, at $q_0$, transitions on the letters 0 and 1 (legal, since cell 0 of a start tape always holds #), the theory with free-$z$ output rules proves both $R(t_x)$ and $\neg R(t_x)$ for every $x$. A computation confirms this on three machines with such transitions (checked derivations of $R(t_{0110})$ and $\neg R(t_{0110})$), and confirms the exact rules on the same machines. In Lemma A.13 phase cert consumes $z$, so the free-$z$ output rules are correct there.

**Relativisation to** Std **is needed.** In $\mathfrak{U}$ every element is in the range of $\mathtt{s}_0$ and $\mathtt{s}_0$ is not injective, so unrelativised patterns $uZ_l$ match pairs, and $u\,\mathrm{pr}(a,b)$ can be a word with the wrong prefix. A computation on the backward theory of a decider for the words with as many 0s as 1s, with $\mathsf{C}(a,b)$ read as $R(\mathrm{pr}(a,b))$ in $\mathfrak{U}$, finds all 3017 instances at word values true and 318 of 12519 instances at pair values false. So without Std the structure $\mathfrak{U}$ is not a model; whether the unrelativised theory is consistent we have not examined.

**Computations.** For that decider (140 rules) and a verifier whose accepting certificate is a position (274 rules), the compilation has no non-linear rule, no overlapping current patterns (checked symbolically, on all pairs of words) and no rule in a halting state, and the runs from all start pairs checked ($|x| \le 10$, $|z| \le 3$, respectively $|x| \le 6$, $|z| \le 5$) end as specified. All 511 data with $|w| \le 8$ have derivations from the relation version (144 sentences) and the observational version (147 sentences), checked by an independent derivation checker; corrupted derivations and a corrupted machine are rejected. For an alternating machine for the words in which every 1 has a 0 to its right, and three further machines with unreachable transitions out of $q_0$ (one universal and one existential with left moves, one deterministic that writes and reads B), all data with $|x| \le 6$ have checked derivations in both versions, and the largest line grows linearly in $|x|$, as reading (a1) requires. An independent implementation agrees.

## A.5 Vocabulary or ontology

**Residuals and the relational form.** For $X \subseteq \{0,1\}^*$ and a word $v$, $X_v := \{u : uv \in X\}$. In the term model $R(t_{uv}) = R(u\,t_v)$, so $X_v$ is what the subtree below $t_v$ sees. $X$ is regular iff it has finitely many residuals: the $X_v$ are the right quotients of $X$, that is, the reversals of the left quotients of the reversal of $X$, and the Myhill–Nerode theorem applies to the reversal (standard).

Let $\mathcal{L}^{\mathrm{rel}}$ have a unary Root, binary $\mathrm{Gr}_0, \mathrm{Gr}_1$, unary $R, P_1, \ldots, P_k$ and unary $\mathrm{Cst}_1, \ldots, \mathrm{Cst}_j$. A structure $\mathfrak{A}$ for $\mathcal{L}_O \cup \{P_i\} \cup \{\mathtt{c}_i\}$ gives $\mathfrak{A}^{\mathrm{rel}}$ with Root $= \{\mathtt{e}\}$, $\mathrm{Gr}_a$ the graph of $\mathtt{s}_a$ ($\mathrm{Gr}_a(x,y)$ iff $\mathtt{s}_a x = y$) and $\mathrm{Cst}_i = \{\mathtt{c}_i\}$; every sentence $\psi$ has a relational translation $\psi^{\mathrm{rel}}$, obtained by unnesting terms, with $\mathfrak{A} \models \psi$ iff $\mathfrak{A}^{\mathrm{rel}} \models \psi^{\mathrm{rel}}$ (standard). $\equiv_q$ is agreement on all sentences of quantifier rank at most $q$. Over a finite relational signature there are finitely many $\equiv_q$-classes, and $\equiv_q$ holds iff the duplicator wins the $q$-round Ehrenfeucht–Fraïssé game (Ehrenfeucht, 1961; Fraïssé, 1954); see Ebbinghaus and Flum (1999) for the textbook form. In Theorem A.16, $\mathrm{ncl}_q$ is the number of $\equiv_q$-classes of $\mathcal{L}^{\mathrm{rel}} \cup \{\mathrm{Rt}, \mathrm{Att}\}$-structures, Rt and Att unary.

**Lemma A.15** (Composition)**.** *(a) For relational structures in one finite signature, $\mathfrak{A}_1 \equiv_q \mathfrak{B}_1$ and $\mathfrak{A}_2 \equiv_q \mathfrak{B}_2$ imply $\mathfrak{A}_1 \sqcup \mathfrak{A}_2 \equiv_q \mathfrak{B}_1 \sqcup \mathfrak{B}_2$, where $\sqcup$ is the disjoint union with each relation interpreted as the union. (b) If $\mathfrak{A}, \mathfrak{B}$ arise from $\mathfrak{A}', \mathfrak{B}'$ (on the same domains) by defining each relation by a quantifier-free formula, then $\mathfrak{A}' \equiv_q \mathfrak{B}'$ implies $\mathfrak{A} \equiv_q \mathfrak{B}$.*

*Proof.* (a) The duplicator answers a move in a component with her winning strategy for that component's game. Within a component the chosen tuples form a partial isomorphism; no relation holds across components on either side, and equality across components fails on both sides. (b) Substituting the defining formulas turns a rank-$q$ sentence about $\mathfrak{A}$ into a rank-$q$ sentence about $\mathfrak{A}'$. □

**Theorem A.16** (Unary theoretical vocabulary sees only regular regularities)**.** *Let $B = \emptyset$, and let $T$ be a finite set of sentences in $\mathcal{L}_O$ plus unary predicate symbols $P_1, \ldots, P_k$ and constants $\mathtt{c}_1, \ldots, \mathtt{c}_j$ ($k, j \geq 0$), compatible with the named structure. If $T \vdash \varphi_w^{\mathbf{1}[w \in X]}$ for every word $w$, then $X$ is regular. Quantitatively, $X$ has at most $\mathrm{ncl}_q + |\mathrm{Suf_c}|$ residuals, where $q$ is the quantifier rank of the relational translation of $\bigwedge T$, $\mathrm{ncl}_q$ the number of classes of rank-$q$ elementary equivalence in the relational signature defined above, and $\mathrm{Suf_c}$ the set of suffixes of the words $w_i$ with $\mathtt{c}_i = t_{w_i}$ in a fixed model of $T \cup \mathrm{Th}_{\mathfrak{T}}$, together with the empty word.*

In outline: by $\mathrm{Th}_{\mathfrak{T}}$ a single edge joins the subtree below $t_v$ to the rest of a model, and unary symbols are labels; so by the composition lemma (Lemma A.15) the labels of one subtree can be replaced by those of another of the same rank-$q$ type without changing the truth of $T$. As $T$ proves the literals, the two residuals coincide.

*Proof.* Take $\mathfrak{A} \models T \cup \mathrm{Th}_{\mathfrak{T}}$. The closed terms denote pairwise distinct elements, and $R^{\mathfrak{A}}(t_w) = \mathbf{1}[w \in X]$ because $T$ proves the literals. For a word $v \neq \varepsilon$, $v = a\hat{v}$ with $a$ a letter, let $\mathrm{Sub}_v := \{t_{uv} : u \in \{0,1\}^*\}$ (standard elements), let $\mathfrak{S}_v$ be the restriction of $\mathfrak{A}^{\mathrm{rel}}$ to $\mathrm{Sub}_v$ with a new unary $\mathrm{Rt} := \{t_v\}$ (and $\mathrm{Att} := \emptyset$), and let $\mathfrak{C}_v$ be the restriction to the rest of the domain with a new unary $\mathrm{Att} := \{t_{\hat{v}}\}$ (and $\mathrm{Rt} := \emptyset$).

*One edge between the parts.* $\mathrm{Sub}_v$ is closed under $\mathtt{s}_0, \mathtt{s}_1$, and $\mathtt{e} \notin \mathrm{Sub}_v$, as $\mathtt{e}$ is not in the range of $\mathtt{s}_0, \mathtt{s}_1$. If $\mathtt{s}_b y = t_{uv}$ with $y \notin \mathrm{Sub}_v$, then $u = \varepsilon$: otherwise $u = b'u'$ and $t_{uv} = \mathtt{s}_{b'} t_{u'v}$, so $b = b'$ by disjoint ranges and $y = t_{u'v} \in \mathrm{Sub}_v$ by injectivity, both in $\mathrm{Th}_{\mathfrak{T}}$. So $\mathtt{s}_b y = t_v = \mathtt{s}_a t_{\hat{v}}$, hence $b = a$ and $y = t_{\hat{v}}$. Hence $\mathfrak{A}^{\mathrm{rel}}$ is obtained from $\mathfrak{C}_v \sqcup \mathfrak{S}_v$ by redefining $\mathrm{Gr}_a(x,y)$ as $\mathrm{Gr}_a(x,y) \vee (\mathrm{Att}(x) \wedge \mathrm{Rt}(y))$ and forgetting Att and Rt, a quantifier-free redefinition.

*The subtrees are alike up to labels.* For $v, v' \notin \mathrm{Suf_c}$ the map $t_{uv} \mapsto t_{uv'}$ is an isomorphism of the $\{\mathrm{Root}, \mathrm{Gr}_0, \mathrm{Gr}_1, \mathrm{Rt}, \mathrm{Cst}_1, \ldots, \mathrm{Cst}_j\}$-reducts of $\mathfrak{S}_v$ and $\mathfrak{S}_{v'}$: Root and the $\mathrm{Cst}_i$ are empty on both, because a constant lies in $\mathrm{Sub}_v$ only if it is some $t_{w_i}$ with $v$ a suffix of $w_i$, and nonstandard denotations lie in no $\mathrm{Sub}_v$.

*Transplanting.* Suppose $X$ has more than $\mathrm{ncl}_q + |\mathrm{Suf_c}|$ residuals. Choose one word per residual; more than $\mathrm{ncl}_q$ of them lie outside $\mathrm{Suf_c}$, so two of them, $v \neq v'$, have $X_v \neq X_{v'}$ and $\mathfrak{S}_v \equiv_q \mathfrak{S}_{v'}$.

Let $\mathfrak{A}^*$ be $\mathfrak{A}$ with the labels on $\mathrm{Sub}_v$ replaced by those copied from $\mathrm{Sub}_{v'}$: $R^*(t_{uv}) := R^{\mathfrak{A}}(t_{uv'})$ and $P_i^*(t_{uv}) := P_i^{\mathfrak{A}}(t_{uv'})$, everything else unchanged (in particular the $\{\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1\}$-reduct and the constants). Then $\mathfrak{S}_v^* \cong \mathfrak{S}_{v'} \equiv_q \mathfrak{S}_v$ and $\mathfrak{C}_v^* = \mathfrak{C}_v$, so by Lemma A.15, $(\mathfrak{A}^*)^{\mathrm{rel}} \equiv_q \mathfrak{A}^{\mathrm{rel}}$ and $\mathfrak{A}^* \models T$. Its reduct still satisfies $\mathrm{Th}_{\mathfrak{T}}$, so, as $T$ proves the literals, $R^*(t_{uv}) = \mathbf{1}[uv \in X]$ for all $u$. But $R^*(t_{uv}) = \mathbf{1}[uv' \in X]$. Hence $X_v = X_{v'}$, a contradiction. Overlapping subtrees, $\mathrm{Sub}_v \supset \mathrm{Sub}_{v'}$, are harmless, since the labels are copied from the original structure. □

Unary predicates and constants are labels: they do not connect a subtree to its context, so the composition lemma applies. A binary theoretical relation can connect any two subtrees (Theorem A.14(i)); objects that violate $\mathrm{Th}_{\mathfrak{T}}$ (pairs, with $\mathtt{s}_1\mathtt{s}_1 z = z$) can carry information between subtrees through $\mathtt{s}_0$, which is not injective on them (Theorem A.14(ii)); and a finite set is essential, since infinite sets of padded literals are true of the named objects and fit $\mathsf{P}$ (Theorem A.6). The proof uses no tree automata and allows nonstandard elements and constants. Theorem A.16 is in the spirit of the reported example of Craig and Vaught that unary additional predicates do not suffice in general, in a different setting; we have not established that it is new.

**Theorem A.17** (Every regular regularity has a short observational theory)**.** *For every regular $X$ there is a finite $T \subseteq \mathrm{Sent}_{\mathcal{L}_O}$, true in $(\mathfrak{T}, X)$, such that every datum has a $\mathsf{K}$-derivation from $T$ with $O(|w|)$ lines, each of size $O(|w|)$, the constants depending on $X$. So the budgets needed are linear under (a1); constant under (a2) and (b2) (at least the largest member, respectively the total size of $T$; with smaller budgets, a finite table of short data as in Theorem A.14(iv)); and quadratic under (b).*

*Proof.* Let $k_X$ be the number of residuals of $X$, and write $A/a := \{u : ua \in A\}$, so $X_v/a = X_{av}$. Let $\approx_j$ be the equivalence "agree on all words of length less than $j$" on the finite set $\{X_v\}$. Then $A \approx_{j+1} A'$ iff $(\varepsilon \in A \Leftrightarrow \varepsilon \in A')$ and $A/a \approx_j A'/a$ for $a = 0, 1$, so $\approx_{j+1}$ is a fixed function of $\approx_j$; once $\approx_j = \approx_{j+1}$ the sequence is constant; $\approx_0$ has one class and each strict refinement adds a class; so $\approx_{k_X - 1}$ is equality. Hence the window $\{0,1\}^{<\mathrm{wd}_X}$ with $\mathrm{wd}_X := \max(k_X - 1, 1)$, or any smaller window that contains $\varepsilon$ and separates the residuals, determines $X_v$. Put $\mathrm{sig}(v) := X_v \cap \{0,1\}^{<\mathrm{wd}_X}$; then $\mathrm{sig}(av)$ is a function $\sigma \mapsto \sigma_a$ of $\mathrm{sig}(v)$. With $\mathrm{Sg}_\sigma(x) := \bigwedge_{|u| < \mathrm{wd}_X} R(ux)^{\sigma(u)}$ let

$$T := \{\mathrm{Sg}_{\mathrm{sig}(\varepsilon)}(\mathtt{e})\} \cup \{\forall x\, (\mathrm{Sg}_\sigma(x) \to \mathrm{Sg}_{\sigma_a}(\mathtt{s}_a x)) : \sigma \text{ realised},\ a \in \{0,1\}\}.$$

$\mathrm{Sg}_\sigma(t_v)$ holds in $(\mathfrak{T}, X)$ iff $\mathrm{sig}(v) = \sigma$, so $T$ is true there. Derive $\mathrm{Sg}_{\mathrm{sig}(v)}(t_v)$ for the suffixes $v$ of $w$ from the inside out (one A4 instance and modus ponens steps per letter) and read off the conjunct $u = \varepsilon$, which is $R(t_w)^{\mathbf{1}[w \in X]}$. In a curried form with one literal per conclusion only A4 and modus ponens occur. □

The bound $\max(k_X - 1, 1)$ on the window is attained: on random automata the least separating window equals $k_X - 1$ in many cases (90 of 1500 random automata with $k_X \geq 3$, and never above $\max(k_X - 1, 1)$; a computation, which also checks the theories of two regular languages on all data with $|w| \leq 9$, and three more, in an independent implementation, on all data with $|w| \leq 8$).

**Corollary A.18** (Vocabulary or ontology)**.** *On literal sequences with $B = \emptyset$ (membership is linear-time for finite sets), the fit classes of finite theories are:*

| *finite theories* | *(a2), (b2)*$^\dagger$ | *(b): polynomial* | *(a1): polynomial* |
|---|---|---|---|
| $\mathcal{L}_O$ *+ unary predicates + constants, compatible with the named structure* | *regular* | *regular* | *regular* |
| $\mathcal{L}_O$ *+ one binary relation, about the named objects* | *all decidable* | $\mathsf{NP} \cap \mathsf{coNP}$ | EXPTIME |
| $\mathcal{L}_O$ *only, quantifying over unnamed objects* | *all decidable* | $\mathsf{NP} \cap \mathsf{coNP}$ | EXPTIME |
| *all finite theories, any vocabulary* | *all decidable* | $\mathsf{NP} \cap \mathsf{coNP}$ | EXPTIME |

(†*at every nondecreasing unbounded budget with* $h(m) \geq m+1$*.) The theories of row 3 are inconsistent with* $\mathrm{Th}_{\mathfrak{T}}$ *(Theorem A.16).*

*Proof.* Row 1: Theorem A.16 gives the upper bound (at any budget, and also with free deduction) and Theorem A.17 the lower bound. Rows 2 and 3: Theorem A.14. Row 4: the upper bounds are those of all hypotheses (**??**), the lower bounds rows 2 and 3. □

**Corollary A.19** (With free deduction too)**.** *Let* $B = \emptyset$*. For a decidable non-regular* $X$*, let* $K^{\mathrm{named}}_{\mathrm{fin}}(D)$ *be the least* $K(\psi)$ *over single sentences* $\psi$ *in* $\mathcal{L}_O$ *plus any finitely many unary predicates and constants that are compatible with the named structure and prove* $\psi_D$*. Then* $K^{\mathrm{named}}_{\mathrm{fin}}(D^X_n) \to \infty$*, while* $K^{\mathcal{L}_O \cup \{\mathsf{C}\}}_{\mathrm{fin}}(D^X_n) \leq K(\mathcal{V}) + c$ *and* $K^{\mathcal{L}_O}_{\mathrm{fin}}(D^X_n) \leq K(\mathcal{V}) + c$ *for a decider* $\mathcal{V}$ *of* $X$*.*

*Proof.* By Theorem A.16 no admissible sentence proves all of $D^X$; each fails some datum, and only finitely many sentences have $K \leq j$. The upper bounds are given by the conjunctions of $T^{\mathsf{C}}_{\mathcal{V}}$ and $T^{O}_{\mathcal{V}}$ (Theorem A.14), computable from $\mathcal{V}$. □

The upper bound with $\mathsf{C}$ is an effective instance of the theorem of Kleene and of Craig and Vaught (Kleene, 1952; Craig and Vaught, 1958), as stated in Pakhomov and Visser (2019): the axioms of $\mathrm{Th}_{\mathfrak{T}}$ listed after Definition A.1 together with $\Gamma_{f_X}$ form an r.e. theory without finite models, which therefore has a finitely axiomatised extension with one additional binary relation symbol into which every model of it, $(\mathfrak{T}, X)$ included, expands.

**Proposition A.20** (A forced ontological commitment)**.** *Let* $B = \emptyset$*. (a) A finite theory in* $\mathcal{L}_O$ *plus unary predicates and constants that proves every literal datum of a non-regular* $X$ *proves some* $\{\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1\}$*-sentence false in the named structure. (b) For a verifier* $\mathcal{V}$*,* $T^O_{\mathcal{V}} \vdash \exists z\, \mathsf{s}_1\mathsf{s}_1 z = z$*, and* $T^O_{\mathcal{V}} \vdash \neg\, \mathsf{s}_1\mathsf{s}_1 t_w = t_w$ *for every word* $w$*. So* $T^O_{\mathcal{V}} \cup \mathrm{Th}_{\mathfrak{T}}$ *is inconsistent:* $T^O_{\mathcal{V}}$ *is consistent with all the data but proves an observational sentence that is false of every named object and refutable from the observational theory of the named structure. In particular* $T^O_{\mathcal{V}}$ *is not conservative over* $\Gamma_{f_{\mathcal{V}}}$*, although* $\Gamma_{f_{\mathcal{V}}} \cup \mathrm{Th}_{\mathfrak{T}}$ *is consistent.*

*Proof.* (a) By Theorem A.16 the theory is inconsistent with $\mathrm{Th}_{\mathfrak{T}}$, so by compactness it proves the negation of some $\{\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1\}$-sentence true in $\mathfrak{T}$. (b) In a model of $T^O_{\mathcal{V}} \cup \{\forall z\, \neg\, \mathsf{s}_1\mathsf{s}_1 z = z\}$, Std holds everywhere and every $\theta_2(a, b)$ holds vacuously, so the two output rules give $\forall x\, R(x)$ and $\forall x\, \neg R(x)$; no such model exists. $\mathrm{Std}(t_w)$ follows from $\mathrm{Std}(\mathsf{e})$ and the propagation axioms, and $\forall z\, \neg\, \mathsf{s}_1\mathsf{s}_1 z = z \in \mathrm{Th}_{\mathfrak{T}}$. Finally $(\mathfrak{T}, X_{\mathrm{acc}}) \models \mathrm{Th}_{\mathfrak{T}} \cup \Gamma_{f_{\mathcal{V}}}$. □

A computation confirms that without pairs the two output rules of the observational theory cannot both hold (for 200 of 200 random interpretations of $R$).

**Proposition A.21** (Conservativity of the relation version)**.** *Let* $B = \emptyset$*. For a verifier* $\mathcal{V}$*,* $T^{\mathsf{C}}_{\mathcal{V}}$ *is conservative over* $\mathrm{Th}_{\mathfrak{T}} \cup \Gamma_{f_{\mathcal{V}}}$*: for every* $\mathcal{L}_O$*-sentence* $\chi$*,* $T^{\mathsf{C}}_{\mathcal{V}} \cup \mathrm{Th}_{\mathfrak{T}} \vdash \chi$ *iff* $\mathrm{Th}_{\mathfrak{T}} \cup \Gamma_{f_{\mathcal{V}}} \vdash \chi$*.*

*Proof.* ($\Leftarrow$) $T_{\mathcal{V}}^{\mathsf{C}} \vdash \Gamma_{f_{\mathcal{V}}}$ by Theorem A.14. ($\Rightarrow$) Every model $\mathfrak{M}$ of $\mathrm{Th}_{\mathfrak{T}} \cup \Gamma_{f_{\mathcal{V}}}$ expands to a model of $T_{\mathcal{V}}^{\mathsf{C}}$. Let $\mathsf{C}^{\mathfrak{M}}$ be the least subset of $M^2$ that contains the pairs with a halting prefix and is closed under the backward rules, interpreted in $\mathfrak{M}$. The rules and halting axioms hold by construction. In $\mathfrak{M}$ the current patterns are still pairwise disjoint and the values $(Z_r, Z_l)$ of a match are unique, since disjointness and injectivity are first-order consequences of $\mathrm{Th}_{\mathfrak{T}}$ (for instance $p\,Z \neq p'\,Z'$ for incomparable words $p, p'$); so, as in Theorem A.14(i), $\mathsf{C}^{\mathfrak{M}}$ is the set of pairs whose deterministic run in $\mathfrak{M}$ reaches a halting prefix. Consider a start pair $(x, \mathrm{tg}\,\mathrm{st}(\mathrm{start})\,z)$. If $z$ is nonstandard, phase cert never ends: a nonstandard $z \neq \mathtt{e}$ is $\mathtt{s}_a z'$ for some letter $a$ (a sentence of $\mathrm{Th}_{\mathfrak{T}}$) with $z'$ nonstandard, so each step leaves a nonstandard remainder. If $z$ is standard and $x$ nonstandard, phase inp never ends, for the same reason. If both are standard, the run stays in the standard part, a copy of $\mathfrak{T}$, and is the run of Lemma A.13. So $(x, 0\,\mathrm{st}(\mathrm{start})z) \in \mathsf{C}^{\mathfrak{M}}$ implies $x = t_w$ with $\mathcal{V}(w, z') = \mathrm{acc}$ for some $z'$, hence $\mathfrak{M} \models R(t_w)$ by $\Gamma_{f_{\mathcal{V}}}$; likewise for rejection. Completeness gives the claim. □

The design of Lemma A.13, which reads both raw words to their ends before the machine starts, is what makes this work. Proposition A.21 is the conservative-extension form of the theorem of Kleene and of Craig and Vaught for this particular theory, with a derivation-bounded extension.

**Proposition A.22** (Richer observational languages)**.** *(a) Let $B = \emptyset$, and let the observational language have constants $\mathtt{e}_1, \ldots, \mathtt{e}_{r_0}$, unary function symbols $\mathtt{f}_1, \ldots, \mathtt{f}_{r_1}$ and unary observable predicates $R_1, \ldots, R_{r_2}$, the named structure being the free term algebra $\mathfrak{T}'$ on these constants and function symbols and the data all literals $R_i(\tau)^{\pm}$, $\tau$ closed. If a finite theory in this language plus unary theoretical predicates and constants is compatible with $\mathrm{Th}(\mathfrak{T}')$ and proves every datum, then each $R_i$ has finitely many residuals $\{u \in \{1, \ldots, r_1\}^* : R_i(\mathtt{f}_u\tau)\}$ ($\tau$ closed); in particular $\{u : R_i(\mathtt{f}_u\mathtt{e}_l)\}$ is regular for each root $\mathtt{e}_l$. (b) Let the observational language be $\{\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1, R, J\}$ with $J$ a binary observable relation, the data being all literals of $R$ and $J$ on closed terms, and let the world be $J := \{(t_u, t_v) : \#_0(u) = \#_1(v)\}$, $R := \mathrm{EQ} := \{w : \#_0(w) = \#_1(w)\}$, which is not regular. The 13 observational sentences $J(\mathtt{e}, \mathtt{e})$; $\forall y\, \neg J(\mathtt{e}, \mathtt{s}_1 y)$; $\forall x\, \neg J(\mathtt{s}_0 x, \mathtt{e})$; the four sentences $\forall y\, (J(\mathtt{e}, y) \to J(\mathtt{e}, \mathtt{s}_0 y))$, $\forall x \forall y\, (J(x, y) \to J(\mathtt{s}_0 x, \mathtt{s}_1 y))$, $\forall x \forall y\, (J(\mathtt{s}_0 x, y) \to J(\mathtt{s}_0 x, \mathtt{s}_0 y))$, $\forall x \forall y\, (J(x, y) \to J(\mathtt{s}_1 x, y))$, each also with both $J$-atoms negated; and $\forall x\, (J(x, x) \to R(x))$, $\forall x\, (\neg J(x, x) \to \neg R(x))$ are true in $(\mathfrak{T}, R, J)$, so they form a theory about the named objects with no theoretical vocabulary, and they derive every datum with $O(|u| + |v|)$ lines. So with a binary observable, finite observational theories about the named objects fit a non-regular target.*

*Proof.* (a) Use, from $\mathrm{Th}(\mathfrak{T}')$: each $\mathtt{f}_i$ is injective, the ranges of distinct $\mathtt{f}_i$ are disjoint, no root is in any range, and distinct closed terms are distinct. For a non-root closed term $\tau = \mathtt{f}_a\tau'$ put $\mathrm{Sub}_\tau := \{\mathtt{f}_u\tau : u \in \{1, \ldots, r_1\}^*\}$. By these facts an element $y \notin \mathrm{Sub}_\tau$ with $\mathtt{f}_b y \in \mathrm{Sub}_\tau$ must be $\tau'$ with $b = a$ (the one-edge argument in the proof of Theorem A.16, with $\{0, 1\}$ replaced by $\{1, \ldots, r_1\}$), so $\mathfrak{A}^{\mathrm{rel}}$ is $\mathfrak{C}_\tau \sqcup \mathfrak{S}_\tau$ with one quantifier-free redefined edge. The exceptional set is the set of closed subterms of the denotations of the constants, together with the roots (roots are never the top of a $\mathrm{Sub}_\tau$); for $\tau, \tau'$ outside it, $\mathfrak{S}_\tau$ and $\mathfrak{S}_{\tau'}$ are isomorphic up to labels, possibly under different roots. The transplant, which copies all labels $R_1, \ldots, R_{r_2}, P_1, \ldots$ at once, then bounds the number of distinct residuals over all closed terms, for each $R_i$, by $\mathrm{ncl}_q$ plus the size of the exceptional set. (b) Truth: $\#_0$ and $\#_1$ change as the letters say. Every pair $(u, v)$ is $(\varepsilon, \varepsilon)$, $(\varepsilon, 1v')$, $(\varepsilon, 0v')$, $(0u', \varepsilon)$, $(1u', v)$ ($v$ arbitrary, including $\varepsilon$), $(0u', 1v')$ or $(0u', 0v')$, and each case is an instance of a sentence or reduces by one sentence, in its positive or negative form, to a pair of smaller total length (for $(1u', v)$ the sentence $\forall x \forall y\, (J(x, y) \to J(\mathtt{s}_1 x, y))$ at $y := t_v$). The two $R$-sentences finish. □

A computation finds the 13 sentences true for $|x|, |y| \leq 6$, and checked derivations of all 3969 $J$-literals with $|u|, |v| \leq 5$ and all 1023 $R$-literals with $|w| \leq 9$; the windowed residual count of $J$ grows ($5, 7, 9$ at depths $2, 3, 4$).

**Naming composites without observing them.** If the observational language has a binary function symbol $\mathtt{p}$ but the data concern only the words $t_w$, then $T_{\mathcal{V}}^{\mathsf{C}}$ with every atom $\mathsf{C}(a, b)$ replaced by $R(\mathtt{p}(a, b))$, with no relativisation, is about the named objects (the term algebra on $\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1, \mathtt{p}$, with $R(\mathtt{p}(a, b)) := \mathbf{1}[(a, b) \in \mathsf{C}^{\mathfrak{T}}]$ for words $a, b$, false for other composites, and $R := X_{\mathrm{acc}}$ on words), and fits every decidable $X$ under (a2) and (b2). In the free term algebra matching is syntactic, and a start pair with a non-word component gets stuck when phase cert or phase inp reaches the $\mathtt{p}$-term, since no rule matches a term whose head is $\mathtt{p}$; so the output rules hold as in Theorem A.14. Theorem A.16 thus depends on the composites being unnamed, or, if named, observed.

**Question A.23** (Binary observables)**.** What is the fit class of finite observational theories about the named objects when the observational language contains a binary observable relation $J$, as in Proposition A.22(b)? Is it every decidable target? The answer must depend on the observable: with $J = \emptyset$ it is the class of regular sets, by the proof of Theorem A.16.

**Proposition A.24** (Reading (a1) for infinite hypotheses about the named objects)**.** *Let $X \in$ EXPTIME be EXPTIME-hard under polynomial-time many-one reductions. There is a parameter-free set $A \subseteq \mathrm{Sent}_{\mathcal{L}_O}$, true in $(\mathfrak{T}, X)$, with polynomial-time membership, such that every datum $\varphi_w^{\mathbf{1}[w \in X]}$ has a derivation from $A$ by modus ponens and instances of the axiom scheme A1 of $\mathsf{K}$ in which every line has size polynomial in $|w|$. Hence, at polynomial budgets under (a1), the fit class of observational hypotheses about the named objects contains every EXPTIME-hard set in EXPTIME, besides $\mathsf{NP} \cap \mathsf{coNP}$.*

*Proof.* Let $\mathcal{M}$ be an alternating polynomial-space machine for $X$ with all paths halting (Step 1 of the proof of Theorem A.14(iv)), and let $\mathrm{Acc} \in \mathsf{EXPTIME}$ be the set of pairs $(w, \varsigma)$ with $\varsigma$ an accepting configuration of $\mathcal{M}$ on $w$. Let red be a polynomial-time reduction of Acc (pairs coded as words) to $X$; since every configuration accepts or rejects, $\varsigma$ rejects iff $\mathrm{red}(w, \varsigma) \notin X$. For a word $u$ let $\theta_u$ be the valid certificate sentence of **??** encoding $u$, and $\langle \cdot \rangle$ a pairing of words. $A$ consists of, for all $w$ and configurations $\varsigma$ of $\mathcal{M}$ on $w$: $\theta_{\langle w,\varsigma \rangle} \to R(t_{\mathrm{red}(w,\varsigma)})$ for halting accepting $\varsigma$ and $\theta_{\langle w,\varsigma \rangle} \to \neg R(t_{\mathrm{red}(w,\varsigma)})$ for halting rejecting $\varsigma$; at an existential $\varsigma$ with successors $\varsigma_1, \varsigma_2$, $\theta_{\langle w,\varsigma,\varsigma_i \rangle} \to (R(t_{\mathrm{red}(w,\varsigma_i)}) \to R(t_{\mathrm{red}(w,\varsigma)}))$ for $i = 1, 2$ and $\theta_{\langle w,\varsigma,\varsigma_1,\varsigma_2 \rangle} \to (\neg R(t_{\mathrm{red}(w,\varsigma_1)}) \to (\neg R(t_{\mathrm{red}(w,\varsigma_2)}) \to \neg R(t_{\mathrm{red}(w,\varsigma)})))$, and at a universal $\varsigma$ the duals; and $\theta_{\langle w \rangle} \to (R(t_{\mathrm{red}(w,\mathrm{init}(w))}) \to R(t_w))$ and $\theta_{\langle w \rangle} \to (\neg R(t_{\mathrm{red}(w,\mathrm{init}(w))}) \to \neg R(t_w))$. Every member is true in $(\mathfrak{T}, X)$, since $\theta$ is valid and $R(t_{\mathrm{red}(w,\varsigma)})$ says that $\varsigma$ accepts. Membership: read the coded tuple off the $\theta$-part, recompute the red-values and check the transition relation, in polynomial time. A datum is derived along an accepting (rejecting) computation tree: each step cites a member, the valid sentence $\theta_u$ (an A1 instance), and modus ponens; every line has polynomial size, as configurations and red-values do. □

For infinite observational axiom sets true of the named objects, with polynomial membership, the fit class at polynomial budgets is exactly $\mathsf{NP} \cap \mathsf{coNP}$ under (b): certificate sets are true of the named objects, since $\theta_\gamma$ is valid, and the class of all hypotheses bounds it (**??**). Under (a1) it lies between the classes of Proposition A.24 and EXPTIME.

**Question A.25** (Reading (a1) for infinite hypotheses)**.** Is the (a1) fit class of infinite observational hypotheses about the named objects, at polynomial budgets, all of EXPTIME? For an $X$ that is neither easy nor hard, the facts about the named objects that a derivation can use are the values of $X$ itself and sentences about $(\mathfrak{T}, X)$, and it is not clear that they can store an exponential computation. If they cannot, theoretical vocabulary or entities would be needed under (a1) even for infinite hypotheses.

**Question A.26** (A unary theoretical function; soft elimination)**.** (a) Does one unary theoretical function symbol over the named objects suffice for every decidable $X$? A function is not a label, so the proof of Theorem A.16 does not apply; a two-stack configuration cannot be read off a single word by prefix patterns, so the construction of Theorem A.14 does not transfer. (b) Is there a soft version of Proposition A.9, with rate factor 1? Soft axiom induction and soft certificate induction admit no two-sided constant-regret correspondence at any pair of rates (**??**), which suggests that the answer is no.

## A.6 Finite theories with free deduction

In this subsection $B = \emptyset$, and $K^{\mathcal{L}}_{\text{fin}}$, $W^{\mathcal{L}}_{\text{fin}}$ are the finite-axiomatisation versions of **??** in the language $\mathcal{L}$. Recall $K^{\text{pad}}(n) = \min_{m \geq n} K(m)$, which is nondecreasing and unbounded. $\mathcal{L}_{\text{mon}}$, $\mathcal{L}_f$, $\text{Inf}_n$, $D_n$ and $\text{Dinf}_f$ are as in Section 1.5: $\mathcal{L}_{\text{mon}} := \{=\} \cup \{P_i : i \in \mathbb{N}\}$, $\mathcal{L}_f := \mathcal{L}_{\text{mon}} \cup \{f\}$ with $f$ unary, $\text{Inf}_n := \exists x_1 \cdots \exists x_n \bigwedge_{i<j} x_i \neq x_j$, $D_n := \{(\text{Inf}_m, 1) : m \leq n\}$, so that $\psi_{D_n} \equiv \text{Inf}_n$, and $\text{Dinf}_f := \forall x \forall y\, (fx = fy \to x = y) \wedge \exists y \forall x\, fx \neq y$.

**Lemma A.27** (A quantitative finite model property of monadic logic)**.** *If $\psi \in \text{Sent}_{\mathcal{L}_{\text{mon}}}$ uses $k$ predicate symbols, has quantifier rank $q$ and has a model, then it has a model with at most $2^k \max(q, 1)$ elements. For $k = 0$, all sets of size at least $\max(q, 1)$ satisfy the same rank-$q$ sentences.*

*Proof.* Let $\mathfrak{M} \models \psi$ and $q' := \max(q, 1)$. Partition $\mathfrak{M}$ into the $2^k$ cells $\text{Cell}_\epsilon$, $\epsilon \in \{0,1\}^k$, by the values of the $k$ predicates, and let $\mathfrak{M}' \subseteq \mathfrak{M}$ keep $\min(|\text{Cell}_\epsilon|, q')$ elements of each cell; $\mathfrak{M}' \neq \emptyset$. The duplicator wins the $q$-round game on $(\mathfrak{M}, \mathfrak{M}')$, keeping the invariant "chosen pairs lie in the same cell, and $a_i = a_j \iff b_i = b_j$": a repeated element is answered by its partner, and a fresh element in cell $\epsilon$ by a fresh element of that cell on the other side, which exists. If $|\text{Cell}_\epsilon| \geq q'$, both sides have at least $q' \geq q$ elements of the cell and fewer than $q$ are chosen; if $|\text{Cell}_\epsilon| < q'$, both sides have exactly $|\text{Cell}_\epsilon|$ elements of the cell, and the chosen ones are matched bijectively. The invariant gives a partial isomorphism for $=$ and the $P_i$, so $\mathfrak{M}' \equiv_q \mathfrak{M}$ and $\mathfrak{M}' \models \psi$. □

This is a quantitative form of the finite model property of monadic first-order logic, whose decidability goes back to Löwenheim (1915) and Behmann (1922). A computation by exact Ehrenfeucht–Fraïssé games confirms that truncation at $\max(q, 1)$ preserves $\equiv_q$ in all 185 structures tried, that truncation at $q - 1$ breaks it in all 40 tried, and that the bound $2^k q$ is attained.

**Theorem A.28** (Infinity needs a theoretical function)**.** *Let $\text{mb}(\psi) := 2^{k(\psi)} \max(q(\psi), 1)$ for $\psi \in \text{Sent}_{\mathcal{L}_{\text{mon}}}$, with $k(\psi)$ the number of predicate symbols in $\psi$ and $q(\psi)$ its quantifier rank. (a) If $\psi \in \text{Sent}_{\mathcal{L}_{\text{mon}}}$ is consistent and $\psi \vdash \text{Inf}_n$, then $n \leq \text{mb}(\psi)$. No consistent finite monadic theory proves all of $D_\infty$. (b) $K^{\mathcal{L}_{\text{mon}}}_{\text{fin}}(D_n) = K^{\text{pad}}(n) \pm c$. So $K^{\mathcal{L}_{\text{mon}}}_{\text{fin}}(D_n) \to \infty$, and $K^{\mathcal{L}_{\text{mon}}}_{\text{fin}}(D_n) = o(u(n))$ for every computable unbounded nondecreasing $u : \mathbb{N} \to \mathbb{N}$. (c) $\text{Dinf}_f$ is consistent and proves every $\text{Inf}_n$, so $K^{\mathcal{L}_f}_{\text{fin}}(D_n) \leq K(\text{Dinf}_f) + c$. (d) Under the prior $\mathbf{m}$ on $\text{Sent}_{\mathcal{L}_f}$, the posterior probability, given $D_n$, that the finite hypothesis does not use $f$ is at most $2^c \sum_{m \geq n} \mathbf{m}(m) / \mathbf{m}(\text{Dinf}_f)$, which tends to 0. (e) $K^{\{=\}}_A(D_n) \leq c$, by the r.e. theory $\{\text{Inf}_m : m \geq 1\}$.*

*Proof.* (a) By Lemma A.27, $\psi$ has a model with at most $\mathrm{mb}(\psi)$ elements. (b) $\mathrm{mb}(\psi) \geq n$ is computable from $\psi$, so $K(\psi) \geq K(\mathrm{mb}(\psi)) - c \geq K^{\mathrm{pad}}(n) - c$. Conversely, $\mathrm{Inf}_m$ for $m \geq n$ is consistent, proves $\mathrm{Inf}_n$ and has $K(\mathrm{Inf}_m) \leq K(m) + c$. Divergence: $K^{\mathrm{pad}}$ is unbounded because only finitely many $m$ have $K(m) \leq j$. Slow growth: for a computable unbounded nondecreasing $u_1$ let $h_{u_1}(j)$ be the least $n'$ with $u_1(n') \geq j$; then $h_{u_1}(u_1(n)+1) > n$, so $K^{\mathrm{pad}}(n) \leq K(h_{u_1}(u_1(n)+1)) + c \leq 2\log(u_1(n)+1) + c'$. With $u_1 := \lfloor 2^{u/r} \rfloor$ this is at most $2u(n)/r + c''$ for every $r$, hence $o(u)$. (c) $(\mathbb{N}, \text{successor}) \models \mathrm{Dinf}_f$. In any model, with $y_0$ outside the range of $f$, the elements $y_0, fy_0, \ldots, f^{n-1}y_0$ are distinct: $f^i y_0 = f^j y_0$ with $i < j$ gives $y_0 = f^{j-i}y_0$ by injectivity, which lies in the range. (d) The pushforward of $\mathbf{m}$ under the computable partial map $\psi \mapsto \mathrm{mb}(\psi)$ (on $\mathcal{L}_{\mathrm{mon}}$) is at most $2^c\mathbf{m}$; compatible monadic sentences have $\mathrm{mb}(\psi) \geq n$; and the compatible mass with $f$ is at least $\mathbf{m}(\mathrm{Dinf}_f)$. (e) is immediate. □

A computation confirms that $\mathrm{Dinf}_f$ has no model with at most 7 elements.

*Remark* A.29 (The same with named objects). In the language $\{R\} \cup \{c_j : j \in \mathbb{N}\}$ ($R$ unary, $c_j$ constants), with $B = \emptyset$, the data sets $D_n := \{(R(c_j), 1) : j < n\} \cup \{(\exists x\, \neg R(x), 1)\}$ have $K_A(D_n) \leq c$ and $K_{\mathrm{fin}}(D_n) = K^{\mathrm{pad}}(n) \pm c$. So $K_{\mathrm{fin}} - K_A$ is unbounded, and finite axiomatisability is a different simplicity measure.

*Proof.* $T := \{R(c_j) : j \in \mathbb{N}\} \cup \{\exists x\, \neg R(x)\}$ is r.e. with an index of complexity $O(1)$ and is consistent (domain $\mathbb{N} \cup \{*\}$, $c_j \mapsto j$, $R := \mathbb{N}$), so $K_A(D_n) \leq c$. *Upper bound on* $K_{\mathrm{fin}}$: for $N \geq n$ with $K(N) = K^{\mathrm{pad}}(n)$, the sentence $\bigwedge_{j<N} R(c_j) \wedge \exists x\, \neg R(x)$ is consistent, proves $\psi_{D_n}$, and has complexity at most $K(N) + c$. *Lower bound*: let $\psi$ be consistent with $\psi \vdash \psi_{D_n}$. If some $c_j$, $j < n$, does not occur in $\psi$, then $\psi \vdash R(c_j)$ and the theorem on constants give $\psi \vdash \forall x\, R(x)$, which contradicts $\psi \vdash \exists x\, \neg R(x)$ and the consistency of $\psi$. So $\psi$ mentions $c_0, \ldots, c_{n-1}$, and $1 + \max\{j : c_j \text{ occurs in } \psi\} \geq n$ is computable from $\psi$; hence $K(\psi) \geq K^{\mathrm{pad}}(n) - c$. □

**Proposition A.30** (The cost of "at least $n$ things" in symbols and in bits)**.** *Let $|\psi|$ be the number of symbols of $\psi$ (each $P_i$ one symbol), and let $\psi$ range over* consistent *sentences with $\psi \vdash \mathrm{Inf}_n$. (a) In pure equality, $q(\psi) \geq n$ for $n \geq 2$, so $|\psi| \geq n$; $\mathrm{Inf}_n$ itself has $\Theta(n^2)$ symbols. (b) For monadic $\psi$, $|\psi| \geq \log n - \log\log n - O(1)$, and $|\psi| = O(\log n)$ is attained by the* carry counter

$$\mathrm{Cnt}'_k := \forall x\, \mathrm{Cr}_0 x \;\wedge\; \bigwedge_{i<k} \forall x\, (\mathrm{Cr}_{i+1} x \leftrightarrow (\mathrm{Cr}_i x \wedge P_i x)) \;\wedge\; \exists x \bigwedge_{i<k} \neg P_i x \;\wedge\; \forall x \Big( \neg \mathrm{Cr}_k x \to \exists y \bigwedge_{i<k} (P_i y \leftrightarrow (P_i x \oplus \mathrm{Cr}_i x)) \Big),$$

*with $\mathrm{Cr}_0, \ldots, \mathrm{Cr}_k$ further unary predicates, which holds exactly in the monadic structures in which the $\mathrm{Cr}_i$ are the carries of the $P$-vector and all $2^k$ cells of $P_0, \ldots, P_{k-1}$ are nonempty; so $\mathrm{Cnt}'_{\lceil \log n \rceil} \vdash \mathrm{Inf}_n$. (c) With $f$, $O(1)$ symbols suffice ($\mathrm{Dinf}_f$). (d) Code sentences in Polish notation, symbol by symbol, with a prefix-free code on the symbols in which each of the finitely many logical symbols and variables used costs between 1 and $c_0$ bits and $P_i$ costs $\ell(i)$ bits, with $\ell$ the length function of a prefix-free code on $\mathbb{N}$ with $\ell(i) \leq 2\log(i+1) + O(1)$. Then the codes of sentences form a prefix-free set, so $\sum_\psi 2^{-\mathrm{bits}(\psi)} \leq 1$ and $2^{-\alpha\,\mathrm{bits}(\psi)}$, $\alpha \geq 1$, is a sub-probability on sentences. In this code pure equality costs at least $n$ bits, $f$ costs $O(1)$ bits, and monadic sentences cost $\Theta(\log n \cdot \log\log n)$ bits.*

*Proof.* (a) Let $\mathfrak{M} \models \psi$; then $|\mathfrak{M}| \geq n$. If $|\mathfrak{M}| \geq \max(q, 1)$, Lemma A.27 with $k = 0$ makes every set of size $\max(q, 1)$ a model of $\psi$, so $\max(q, 1) \geq n$; otherwise $\max(q, 1) > |\mathfrak{M}| \geq n$. Either way $q \geq n$ when $n \geq 2$, and each quantifier is a symbol. (b) *Lower bound*: Theorem A.28(a) gives $n \leq 2^k \max(q, 1)$ with $k, q \leq |\psi|$. *Correctness of* $\mathrm{Cnt}'_k$: the definitional conjuncts force

$\mathrm{Cr}_i x \leftrightarrow \bigwedge_{j<i} P_j x$, the carry into bit $i$ of the binary increment. If all $P$-vectors are realised, the remaining conjuncts hold, since every non-maximal vector has its increment. Conversely, the vector 0 is realised, and if the vector of value $v < 2^k - 1$ is realised, so is that of $v+1$ ($\neg \mathrm{Cr}_k x$ says that the vector of $x$ is not all ones, and bit $i$ of $v+1$ is $P_i \oplus \mathrm{Cr}_i$); by induction all $2^k$ vectors are realised, so there are at least $2^k$ elements. Each conjunct indexed by $i$ has constant size. (c) Theorem A.28(c). (d) *Prefix-freeness*: a sentence in Polish notation is determined, symbol by symbol, by its own beginning, so no sentence code is a proper prefix of another. *Lower bound, monadic*: a consistent $\psi \vdash \mathrm{Inf}_n$ with $k$ distinct predicate symbols and rank $q$ contains at least $q$ quantifier symbols, of at least one bit each, and $k$ distinct predicate codewords. Sort their lengths $\ell_{(1)} \le \cdots \le \ell_{(k)}$; by Kraft's inequality $j\, 2^{-\ell_{(j)}} \le \sum_{i \le j} 2^{-\ell_{(i)}} \le 1$, so $\ell_{(j)} \ge \log j$ and the predicate codewords cost at least $\log k! \ge k \log k - k \log e$ bits. With $n \le 2^k \max(q, 1)$: if $k \ge \frac{1}{2} \log n$, the cost is at least $\frac{1}{2} \log n \, (\log \log n - 1) - \frac{\log e}{2} \log n$; if $k < \frac{1}{2} \log n$, then $q \ge n/2^k > \sqrt{n}$. Either way $\mathrm{bits}(\psi) = \Omega(\log n \log \log n)$. *Upper bound*: $\mathrm{Cnt}'_k$ with $k = \lceil \log n \rceil$, its $2k+1$ predicates numbered $1, \ldots, 2k+1$, has $O(k)$ symbols of at most $\max(c_0, 2 \log(2k+2) + O(1))$ bits each, so $O(\log n \log \log n)$ bits. *Pure equality*: at least $q \ge n$ symbols of at least one bit. *With $f$*: $\mathrm{Dinf}_f$ is a fixed sentence. □

So the single best theory with $f$ outweighs the best monadic theory by a factor $n^{\Theta(\alpha)}$ and the best pure-equality theory by $2^{\Omega(\alpha n)}$ under the weight $2^{-\alpha|\psi|}$, and by $2^{\Theta(\alpha \log n \log \log n)}$ and $2^{\Omega(\alpha n)}$ under the normalisable weight of (d); with each $P_i$ one symbol, the weight $2^{-\alpha|\psi|}$ is not normalisable, so the first comparison is between single theories, not under a prior. A computation gives $64k + 13$ symbols for $\mathrm{Cnt}'_k$ in one convention (linear in $k$ in three conventions), checks that $\mathrm{Cnt}'_k$ holds iff all cells are nonempty on all 6560 carry-labelled structures with $k = 3$ and cell counts at most 2, and finds the ratios of the lower and upper bit bounds to $\log n \log \log n$ bounded for $n = 2^8, \ldots, 2^{1024}$. The counter is elementary, and we make no novelty claim for it.

**Theorem A.31** (The theorems of PA as data)**.** *Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$ and let $D_\infty$ list the theorems of* PA*, labelled true, in the order of an effective enumeration of* PA*-derivations, so that every axiom of* PA *occurs. (a) $K_{\mathrm{fin}}^{\mathcal{L}_{\mathrm{ar}}}(D_n) \to \infty$ and $W_{\mathrm{fin}}^{\mathcal{L}_{\mathrm{ar}}}(D_n) \to 0$; and $K_{\mathrm{fin}}^{\mathcal{L}_{\mathrm{ar}}}(D_n) \le K(n) + c$. (b) $K_A^{\mathcal{L}_{\mathrm{ar}}}(D_n) \le K(\mathsf{PA}) + c = O(1)$. (c) In the three-sorted extension $\mathcal{L}_{\mathrm{ar}}^+$ of Theorem A.33, $K_{\mathrm{fin}}^{\mathcal{L}_{\mathrm{ar}}^+}(D_n) \le c$, and the posterior over finite $\mathcal{L}_{\mathrm{ar}}^+$-theories puts mass tending to 1 on theories that use the new sorts. The winning theory adds no $\mathcal{L}_{\mathrm{ar}}$-content: its $\mathcal{L}_{\mathrm{ar}}$-consequences are exactly those of* PA*.*

*Proof.* We use the theorem of Ryll-Nardzewski (1952): no consistent extension of PA in $\mathcal{L}_{\mathrm{ar}}$ is finitely axiomatisable; equivalently, no consistent $\mathcal{L}_{\mathrm{ar}}$-sentence proves every axiom of PA. (a) So every consistent $\psi$ fails to prove some axiom $\alpha_\psi$ of PA, which occurs in $D_\infty$, and $\psi$ is incompatible with $D_n$ for large $n$. Only finitely many $\psi$ have $K(\psi) \le j$, which gives $K_{\mathrm{fin}} \to \infty$, and dominated convergence ($\sum \mathbf{m} \le 1$) gives $W_{\mathrm{fin}} \to 0$. The conjunction of the first $n$ data is consistent, given $\mathrm{Con}(\mathsf{PA})$, and computable from $n$. (b) PA is r.e. (c) Theorem A.33 with $\mathsf{W}_e = \mathsf{PA}$ gives a single $\mathcal{L}_{\mathrm{ar}}^+$-sentence $\mathrm{Fin}_e$ whose $\mathcal{L}_{\mathrm{ar}}$-consequences are those of PA; with (a), the posterior mass of the $\mathcal{L}_{\mathrm{ar}}$-sentences tends to 0 while that of $\mathrm{Fin}_e$ is a fixed positive number. □

A classical instance of (c) is $\mathrm{ACA}_0$, in the two-sorted language of second-order arithmetic, which is finitely axiomatisable and has the same first-order consequences as PA (see Simpson, 2009).

*Remark* A.32 (A truth predicate for PA). A theoretical *predicate* on the same domain also appears to give $O(1)$: $\mathrm{I}\Sigma_1$ plus compositional truth clauses for a unary predicate Tr plus $\forall \varphi \, (\text{PA-Ax}(\varphi) \to \mathrm{Tr}(\varphi))$ is a finite theory, true in $\mathbb{N}$, that proves every axiom of PA.

*Proof sketch.* We do not write out the derivation of the Tarski biconditionals from the clauses over $\mathrm{I}\Sigma_1$. For each standard sentence $\alpha$ the theory proves $\mathrm{Tr}(\ulcorner\alpha\urcorner) \leftrightarrow \alpha$ by external induction on $\alpha$ from the compositional clauses, using that $\mathrm{I}\Sigma_1$ proves the needed facts about the coding (that numerals evaluate correctly needs $\Sigma_1$-induction); with $\mathrm{Tr}(\ulcorner\alpha\urcorner)$ for every axiom $\alpha$ of $\mathsf{PA}$ this proves every axiom. $\mathrm{I}\Sigma_1$ is finitely axiomatisable (standard; see Hájek and Pudlák, 1993). So in arithmetic a unary theoretical predicate appears to suffice, because the domain already carries $+$ and $\cdot$, which code syntax; in pure equality unary predicates never help (Theorem A.28(a)), nor over observation sequences (Theorem A.16). We do not claim that unary predicates help only where the domain codes syntax.

**Theorem A.33** (r.e. axiom induction is finite-theory induction with theoretical sorts)**.** *Let $\mathcal{L}$ be a finite language (relation, function and constant symbols, with equality) and $B \subseteq \mathrm{Sent}_{\mathcal{L}}$ decidable. Let $\mathcal{L}^+$ be the three-sorted extension (many-sorted first-order logic, every sort with a nonempty domain) with a sort $\mathsf{o}$ carrying $\mathcal{L}$; a sort $\mathsf{n}$ with $0, S, +, \cdot, <$; and a sort $\mathsf{a}$ of assignments, with* $\mathrm{val} : \mathsf{a} \times \mathsf{n} \to \mathsf{o}$, $\mathrm{upd} : \mathsf{a} \times \mathsf{n} \times \mathsf{o} \to \mathsf{a}$, $\mathrm{den} : \mathsf{n} \times \mathsf{a} \to \mathsf{o}$ *and a relation* $\mathrm{Sat} \subseteq \mathsf{n} \times \mathsf{a}$*. There is a computable injective map $e \mapsto \mathrm{Fin}_e$ from indices to single $\mathcal{L}^+$-sentences such that for every $e$*

$$\{\chi \in \mathrm{Sent}_{\mathcal{L}} : B \cup \{\mathrm{Fin}_e\} \vdash \chi\} = \mathrm{Cn}(\mathsf{W}_e),$$

*both sides being inconsistent together. Consequently, for every finite data set $D$ in $\mathcal{L}$, $K^{\mathcal{L}^+}_{\mathrm{fin}}(D) = K^{\mathcal{L}}_A(D) \pm c$ and $2^{-c}W^{\mathcal{L}}_A(D) \le W^{\mathcal{L}^+}_{\mathrm{fin}}(D) \le 2^c W^{\mathcal{L}}_A(D)$.*

*Proof. Construction.* Fix a Gödel numbering of the $\mathcal{L}$-terms and $\mathcal{L}$-formulas (variables $x_0, x_1, \dots$) and $\Sigma_1$ formulas of $\mathcal{L}_{\mathrm{ar}}$ that define in $\mathbb{N}$ the syntactic relations: $\mathrm{Var}(c, v)$, "$c$ codes $x_v$"; for each function symbol $\mathsf{g}$ of arity $k$ (constants: $k = 0$), $\mathrm{App}_{\mathsf{g}}(c, d_1, \dots, d_k)$, "the $d_i$ code terms $\tau_i$ and $c$ codes $\mathsf{g}(\tau_1, \dots, \tau_k)$"; for each relation symbol $\mathsf{r}$, $\mathrm{At}_{\mathsf{r}}(c, \vec{d})$, and $\mathrm{Eq}(c, d, d')$, for atomic formulas over coded terms; $\mathrm{Neg}(c, d)$, "$d$ codes a formula $\chi$ and $c$ codes $\neg\chi$"; $\mathrm{Imp}(c, d, d')$; $\mathrm{All}(c, v, d)$, "$d$ codes $\chi$ and $c$ codes $\forall x_v \chi$"; and $\mathrm{Enum}_e(c)$, "$c \in \mathsf{W}_e$ and $c$ codes an $\mathcal{L}$-sentence". The bounded quantifiers in these formulas use $<$. $\mathrm{Fin}_e$ is the conjunction of:

- (Ar) Robinson's $\mathsf{Q}$ on the sort $\mathsf{n}$, including the defining axiom $\forall x \forall y\,(x < y \leftrightarrow \exists z\,(Sz + x = y))$, so that $\mathsf{Q}$ proves every true $\Sigma_1$ sentence whose bounded quantifiers use $<$ (standard; see Boolos et al., 2007);
- (Up) $\forall\alpha\forall v\forall x\,\mathrm{val}(\mathrm{upd}(\alpha, v, x), v) = x$ and $\forall\alpha\forall v\forall v'\forall x\,(v \neq v' \to \mathrm{val}(\mathrm{upd}(\alpha, v, x), v') = \mathrm{val}(\alpha, v'))$;
- (Den) $\forall c\forall v\,(\mathrm{Var}(c, v) \to \forall\alpha\,\mathrm{den}(c, \alpha) = \mathrm{val}(\alpha, v))$ and, for each $\mathsf{g}$, $\forall c\forall\vec{d}\,(\mathrm{App}_{\mathsf{g}}(c, \vec{d}) \to \forall\alpha\,\mathrm{den}(c, \alpha) = \mathsf{g}(\mathrm{den}(d_1, \alpha), \dots, \mathrm{den}(d_k, \alpha)))$;
- (Sat) for each $\mathsf{r}$, $\forall c\forall\vec{d}\,(\mathrm{At}_{\mathsf{r}}(c, \vec{d}) \to \forall\alpha\,(\mathrm{Sat}(c, \alpha) \leftrightarrow \mathsf{r}(\mathrm{den}(d_1, \alpha), \dots)))$; the same for Eq with $=$; $\forall c\forall d\,(\mathrm{Neg}(c, d) \to \forall\alpha\,(\mathrm{Sat}(c, \alpha) \leftrightarrow \neg\mathrm{Sat}(d, \alpha)))$; the same for Imp with $\to$; and $\forall c\forall v\forall d\,(\mathrm{All}(c, v, d) \to \forall\alpha\,(\mathrm{Sat}(c, \alpha) \leftrightarrow \forall x\,\mathrm{Sat}(d, \mathrm{upd}(\alpha, v, x))))$, with $x$ of sort $\mathsf{o}$;
- (Ax) $\forall c\,(\mathrm{Enum}_e(c) \to \forall\alpha\,\mathrm{Sat}(c, \alpha))$, with $e$ written as a numeral, so that $e \mapsto \mathrm{Fin}_e$ is injective.

($\supseteq$) Let $\varphi \in \mathsf{W}_e$. $\mathsf{Q}$ proves the true $\Sigma_1$ sentence $\mathrm{Enum}_e(\underline{\ulcorner\varphi\urcorner})$, so $\mathrm{Fin}_e \vdash \forall\alpha\,\mathrm{Sat}(\underline{\ulcorner\varphi\urcorner}, \alpha)$. By external induction on terms $\tau$ and formulas $\chi$ whose variables are among $x_{v_1}, \dots, x_{v_j}$: $\mathrm{Fin}_e \vdash \forall\alpha\,\mathrm{den}(\underline{\ulcorner\tau\urcorner}, \alpha) = \tau^\alpha$ and $\mathrm{Fin}_e \vdash \forall\alpha\,(\mathrm{Sat}(\underline{\ulcorner\chi\urcorner}, \alpha) \leftrightarrow \chi^\alpha)$, where $^\alpha$ replaces each $x_v$ by $\mathrm{val}(\alpha, \underline{v})$. Each step instantiates one clause at the numerals of the immediate subterms or subformulas, whose guard is a true $\Sigma_1$ sentence and hence provable in $\mathsf{Q}$. The quantifier step uses (Up) and $\mathsf{Q} \vdash \underline{v} \neq \underline{v'}$ for $v \neq v'$, which give $\mathrm{val}(\mathrm{upd}(\alpha, \underline{v}, x), \underline{v}) = x$ and $\mathrm{val}(\mathrm{upd}(\alpha, \underline{v}, x), \underline{v'}) = \mathrm{val}(\alpha, \underline{v'})$,

so the induction hypothesis at $\mathrm{upd}(\alpha, \underline{v}, x)$ turns $\forall x\, \mathrm{Sat}(\ulcorner \underline{\chi} \urcorner, \mathrm{upd}(\alpha, \underline{v}, x))$ into $(\forall x_v \chi)^\alpha$. Extra, nonstandard witnesses of the guards in other models only add constraints and cannot block these derivations. For the sentence $\varphi$, $\varphi^\alpha = \varphi$, and the sort $\mathsf{a}$ is nonempty, so $\mathrm{Fin}_e \vdash \varphi$. ($\subseteq$) Let $\mathfrak{M}$ be an $\mathcal{L}$-structure with $\mathfrak{M} \models B \cup \mathsf{W}_e$. Expand it: the sort $\mathsf{n}$ is $\mathbb{N}$, with the true $<$; the sort $\mathsf{a}$ is $M^{\mathbb{N}}$, with $\mathrm{val}(\alpha, v) := \alpha(v)$ and $\mathrm{upd}(\alpha, v, x) := \alpha[v \mapsto x]$; $\mathrm{den}(c, \alpha)$ is the value under $\alpha$ of the term coded by $c$ (a fixed element if $c$ codes no term); and $\mathrm{Sat}(c, \alpha)$ holds iff $c$ codes a formula $\chi$ with $\mathfrak{M} \models \chi[\alpha]$. The guards are evaluated in $\mathbb{N}$, where the $\Sigma_1$ formulas mean what they say, so each clause holds (the guards of Neg, Imp and All require the components to code formulas, so no clause compares a non-formula with a formula), and (Ax) holds because $\mathfrak{M} \models \mathsf{W}_e$. Thus every model of $B \cup \mathsf{W}_e$ expands to a model of $B \cup \{\mathrm{Fin}_e\}$, and by completeness every $\mathcal{L}$-consequence of $B \cup \{\mathrm{Fin}_e\}$ is in $\mathrm{Cn}(\mathsf{W}_e)$. If $B \cup \mathsf{W}_e$ is inconsistent, so is $B \cup \{\mathrm{Fin}_e\}$, by ($\supseteq$); if it is consistent, a model expands, so $B \cup \{\mathrm{Fin}_e\}$ is consistent. *Complexities.* $K(\mathrm{Fin}_e) \leq K(e) + c$, and $\mathrm{Fin}_e$ is compatible with $D$ iff $e$ is. Conversely, the map $\chi \mapsto \mathrm{ct}(\chi)$, an index of $\{\varphi \in \mathrm{Sent}_{\mathcal{L}} : B \cup \{\chi\} \vdash \varphi\}$, is computable; $\mathrm{ct}(\chi)$ is compatible with $D$ iff $\chi$ is (the set contains $\neg \forall x\, x = x$ iff $B \cup \{\chi\}$ is inconsistent); and $K(\mathrm{ct}(\chi)) \leq K(\chi) + c$. For the weights, $\mathbf{m}(\mathrm{Fin}_e) \geq 2^{-c}\mathbf{m}(e)$ because $e \mapsto \mathrm{Fin}_e$ is computable and injective, and $\sum_{\chi : \mathrm{ct}(\chi) = j} \mathbf{m}(\chi) \leq 2^c \mathbf{m}(j)$ because pushforwards of $\mathbf{m}$ under computable maps are dominated by $\mathbf{m}$. □

Theorem A.33 is an effective, many-sorted form of the theorem of Kleene (1952) and Craig and Vaught (1958) (as stated in Pakhomov and Visser, 2019). Using extra sorts to remove the condition that the theory have no finite models is routine; what we add is the Bayesian consequence, which needs an effective construction with an injective map and two-sided complexity bounds. We do not treat the one-sorted version.

*Remark* A.34 (Not one tape). Let $\mathcal{L}$ be finite and satisfy the hypotheses of **??**, and for the universal theory machine of $\mathcal{L}^+$ let $M^{\mathrm{fin}}(D) := \mathbb{U}\{\omega : A_\omega \text{ is finite, } B \cup A_\omega \text{ is consistent and} \vdash \psi_D\}$. On the lottery stream $D_\infty$ of **??**, $M^{\mathrm{fin}}(D_n) \to 0$, while $M_A(D_n)$ is bounded below by a positive constant. So the equivalence of Theorem A.33 is not one-tape: random r.e. theories are strictly stronger than random finite theories with any amount of theoretical apparatus.

*Proof.* $\mathcal{L}^+$ is finite, so it has computable syntax, and for finite $A_\omega$ the $\mathcal{L}$-content $\{\chi \in \mathrm{Sent}_{\mathcal{L}} : B \cup A_\omega \vdash \chi\}$ is r.e. and, if $B \cup A_\omega$ is consistent, consistent. By **??**(b) it does not prove every datum of $D_\infty$. So the events $\{\omega : A_\omega \text{ finite, consistent with } B, \vdash \psi_{D_n}\}$ decrease in $n$ to the empty set, and by continuity from above their measure tends to 0. The lower bound on $M_A(D_n)$ is **??**(a). □

**Proposition A.35** (Infinity: Ramsey, Craig and a theoretical function)**.** *In pure equality: (a) the Ramsey sentence $\exists f\, \mathrm{Dinf}_f$ of the theory $\{\mathrm{Dinf}_f\}$, a single existential second-order sentence with no theoretical constant, has in standard semantics exactly the infinite sets as models; (b) $\{\mathrm{Inf}_m : m \geq 1\}$ has exactly the infinite sets as models, and its consequences form a complete theory; (c) so $\exists f\, \mathrm{Dinf}_f$ and $\{\mathrm{Inf}_m : m \geq 1\}$ (a first-order observational axiomatisation, necessarily infinite by Theorem A.28(a)) have the same first-order $\{=\}$-consequences, and no finite first-order theory in $\{=\}$, nor in any monadic extension, has them; (d) the first-order theory $\{\mathrm{Dinf}_f\}$ in the extended language $\{=, f\}$ is finite and has them as its $\{=\}$-consequences.*

*Proof.* (a) If $f$ is injective and not surjective on $A$, then $A$ is infinite (Theorem A.28(c)); conversely an infinite $A$ has a countable subset $\{a_0, a_1, \dots\}$ (countable choice), and the map $a_i \mapsto a_{i+1}$, the identity elsewhere, is injective and misses $a_0$. (b) The first claim is clear. Completeness: by Lemma A.27 with $k = 0$, all sets of size at least $q$ satisfy the same rank-$q$ sentences, so all infinite sets are elementarily equivalent. (c), (d) The same models give the same first-order consequences; the rest is Theorem A.28(a), (c). □

A computation on 1200 random pure-equality sentences of rank 1 to 4 finds their truth constant on sets of sizes $q$ to $q+2$.

**Proposition A.36** (Factoring an inference graph)**.** *Let $\mathcal{L}$ contain unary predicates $\mathrm{A}_1, \ldots, \mathrm{A}_{n_A}$ and $\mathrm{B}_1, \ldots, \mathrm{B}_{n_B}$; let $T_{\mathrm{dir}} := \{\forall x\,(\mathrm{A}_i x \to \mathrm{B}_j x)\}_{i,j}$ and, in $\mathcal{L} \cup \{\mathrm{Mid}\}$, $T_{\mathrm{Mid}} := \{\forall x\,(\mathrm{A}_i x \to \mathrm{Mid}\,x)\}_i \cup \{\forall x\,(\mathrm{Mid}\,x \to \mathrm{B}_j x)\}_j$. An* atom-to-atom implication *is a sentence $\forall x\,(\mathrm{P}x \to \mathrm{P}'x)$ with $\mathrm{P}, \mathrm{P}'$ distinct unary predicate symbols. (a) $\mathrm{Cn}(T_{\mathrm{Mid}}) \cap \mathrm{Sent}_{\mathcal{L}} = \mathrm{Cn}(T_{\mathrm{dir}})$. (b) Every theory of atom-to-atom implications over $\{\mathrm{A}_i, \mathrm{B}_j\}$ equivalent to $T_{\mathrm{dir}}$ has at least $n_A n_B$ members. (c) In full first-order logic, $T_{\mathrm{dir}}$ is equivalent to the single sentence $\forall x\,((\mathrm{A}_1 x \vee \cdots \vee \mathrm{A}_{n_A} x) \to (\mathrm{B}_1 x \wedge \cdots \wedge \mathrm{B}_{n_B} x))$, of length $O(n_A + n_B)$, so* Mid *saves only a constant factor there. (d) Every theory $S$ of atom-to-atom implications over $\{\mathrm{A}_i, \mathrm{B}_j, \mathrm{Mid}\}$ with $\mathrm{Cn}(S) \cap \mathrm{Sent}_{\mathcal{L}} = \mathrm{Cn}(T_{\mathrm{dir}})$ has at least $\min(n_A n_B, n_A + n_B)$ members, and $T_{\mathrm{dir}}$ or $T_{\mathrm{Mid}}$ attains the bound. So the new predicate shortens the theory iff $(n_A - 1)(n_B - 1) > 1$, by $\max(0, n_A n_B - n_A - n_B)$ members.*

*Proof.* (a) $T_{\mathrm{Mid}} \vdash \mathrm{A}_i \to \mathrm{Mid} \to \mathrm{B}_j$, and every model of $T_{\mathrm{dir}}$ expands to one of $T_{\mathrm{Mid}}$ with $\mathrm{Mid} := \bigcup_i \mathrm{A}_i \subseteq \bigcap_j \mathrm{B}_j$. (b) For a set $S$ of atom-to-atom implications over any set of unary predicates, $S \vdash \forall x\,(\mathrm{P}x \to \mathrm{P}'x)$ iff $\mathrm{P}'$ is reachable from P in the graph $S$: if not, the one-element structure in which exactly the atoms reachable from P hold satisfies $S$, P and $\neg\mathrm{P}'$. In models of $T_{\mathrm{dir}}$ no $\mathrm{A}_i \to \mathrm{A}_{i'}$, $\mathrm{B}_j \to \mathrm{B}_{j'}$ $(i \neq i', j \neq j')$ or $\mathrm{B}_j \to \mathrm{A}_i$ holds (take the one-element structures with only $\mathrm{A}_i$ true together with all B's, respectively only $\mathrm{B}_j$ true). So an equivalent $S$ over $\{\mathrm{A}_i, \mathrm{B}_j\}$ contains only edges $\mathrm{A}_i \to \mathrm{B}_j$, and reachability from $\mathrm{A}_i$ to $\mathrm{B}_j$ then needs the edge itself: all $n_A n_B$ of them. (c) Propositional logic. (d) $S$ proves every $\mathrm{A}_i \to \mathrm{B}_j$ and, by the equality of $\mathcal{L}$-consequences, none of the implications excluded in (b). Take a shortest path from $\mathrm{A}_i$ to $\mathrm{B}_j$ in the graph $S$; it has no repeated vertex. An interior vertex $\mathrm{A}_{i'}$ would give $S \vdash \mathrm{A}_i \to \mathrm{A}_{i'}$ $(i' \neq i)$, and an interior $\mathrm{B}_{j'}$ would give $S \vdash \mathrm{B}_{j'} \to \mathrm{B}_j$ $(j' \neq j)$. So the path is the edge $\mathrm{A}_i \to \mathrm{B}_j$ or the path $\mathrm{A}_i \to \mathrm{Mid} \to \mathrm{B}_j$. With $I_A := \{i : (\mathrm{A}_i \to \mathrm{Mid}) \in S\}$ and $I_B := \{j : (\mathrm{Mid} \to \mathrm{B}_j) \in S\}$, every pair outside $I_A \times I_B$ needs its own edge, so $|S| \geq |I_A| + |I_B| + n_A n_B - |I_A||I_B|$. This function is affine in each of $|I_A|$ and $|I_B|$, so its minimum over $[0, n_A] \times [0, n_B]$ is at a corner: $n_A n_B$, $n_A n_B + n_A$, $n_A n_B + n_B$ or $n_A + n_B$. Finally $n_A n_B - (n_A + n_B) = (n_A - 1)(n_B - 1) - 1$. □

A computation that decides the consequence relation exactly (a set of atom-to-atom implications says that the type of every element lies in a set of types, and the $\mathcal{L}$-reducts of its models are the structures whose types lie in the projection of that set) confirms the least sizes $n_A n_B$ without Mid for the nine pairs with $n_A n_B \leq 6$, and $\min(n_A n_B, n_A + n_B)$ with it for all ten pairs up to $(3, 3)$; Mid saves nothing except for $(2, 3)$, $(3, 2)$ (one member) and $(3, 3)$ (three members).

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